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2	Update Content	26	MIPI DSI SWITCH
3	Block Diagram	27	MIPI DSI CONNECTOR
4	Power Tree	28	MIPI DSI Serializer
5-9	Power Supply	29	Hardware Configuration
10	Power&GND of SOC	30	Button & Light sensor
11	DDRC of SOC	31	Installation
12	LPDDR4/4X	32	
13	SYS of SOC	33	MCU FS32K144
14	Peripheral0 of SOC	34	CAN & ISOLATION
15	Peripheral1 of SOC	35	Motor & GPIO
16	Peripheral2 of SOC	36	Battery in
17	JTAG&RESET		
18	SPI FLASH&EMMC		
19	ETH		
20	CODEC		
21	Micro-SD Card		
22	USB		
23	UART		
24	DeSerializer		

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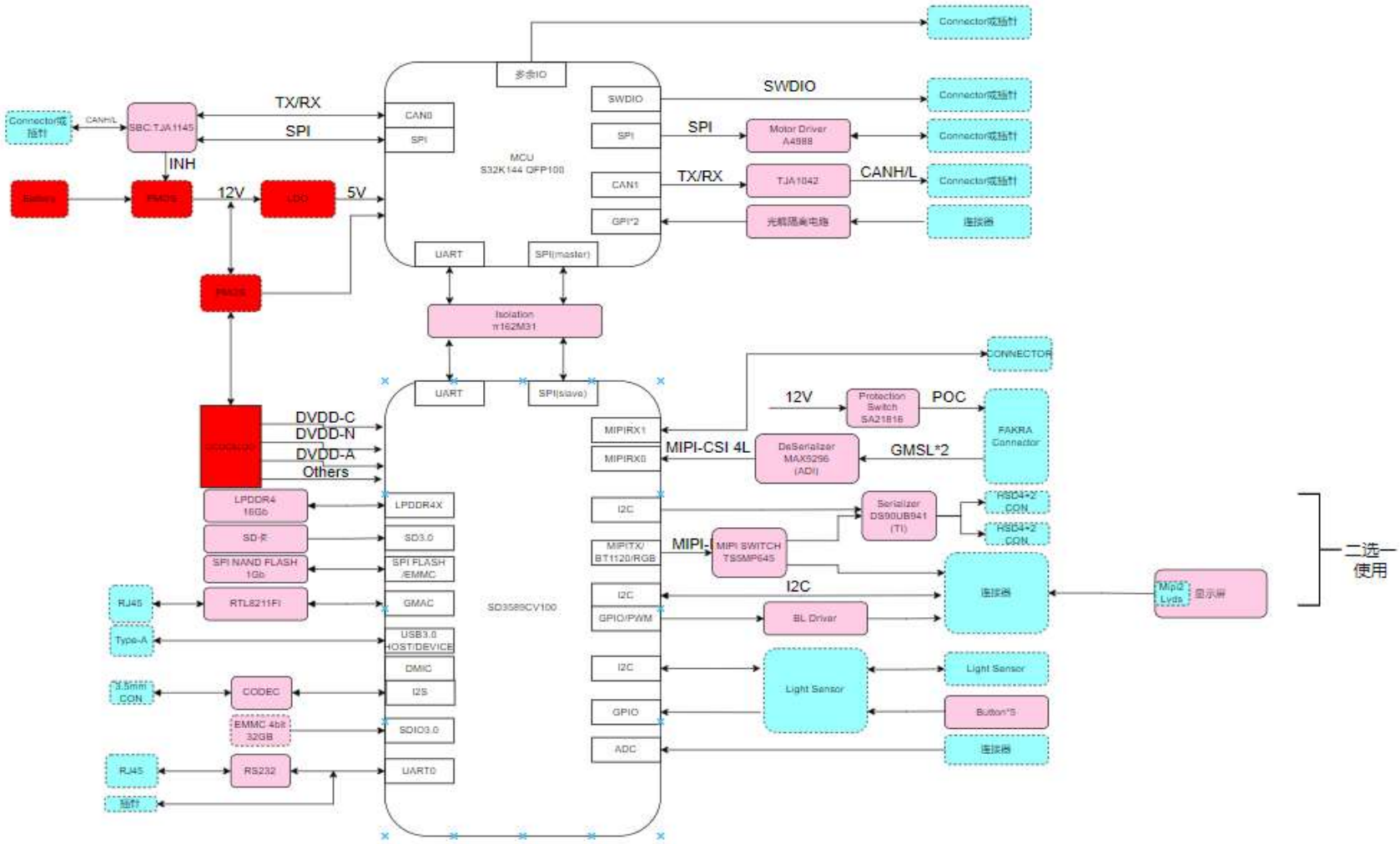
Update Content

VER.	Designer	Date	Description
VER.A		2024-10-11	Initial

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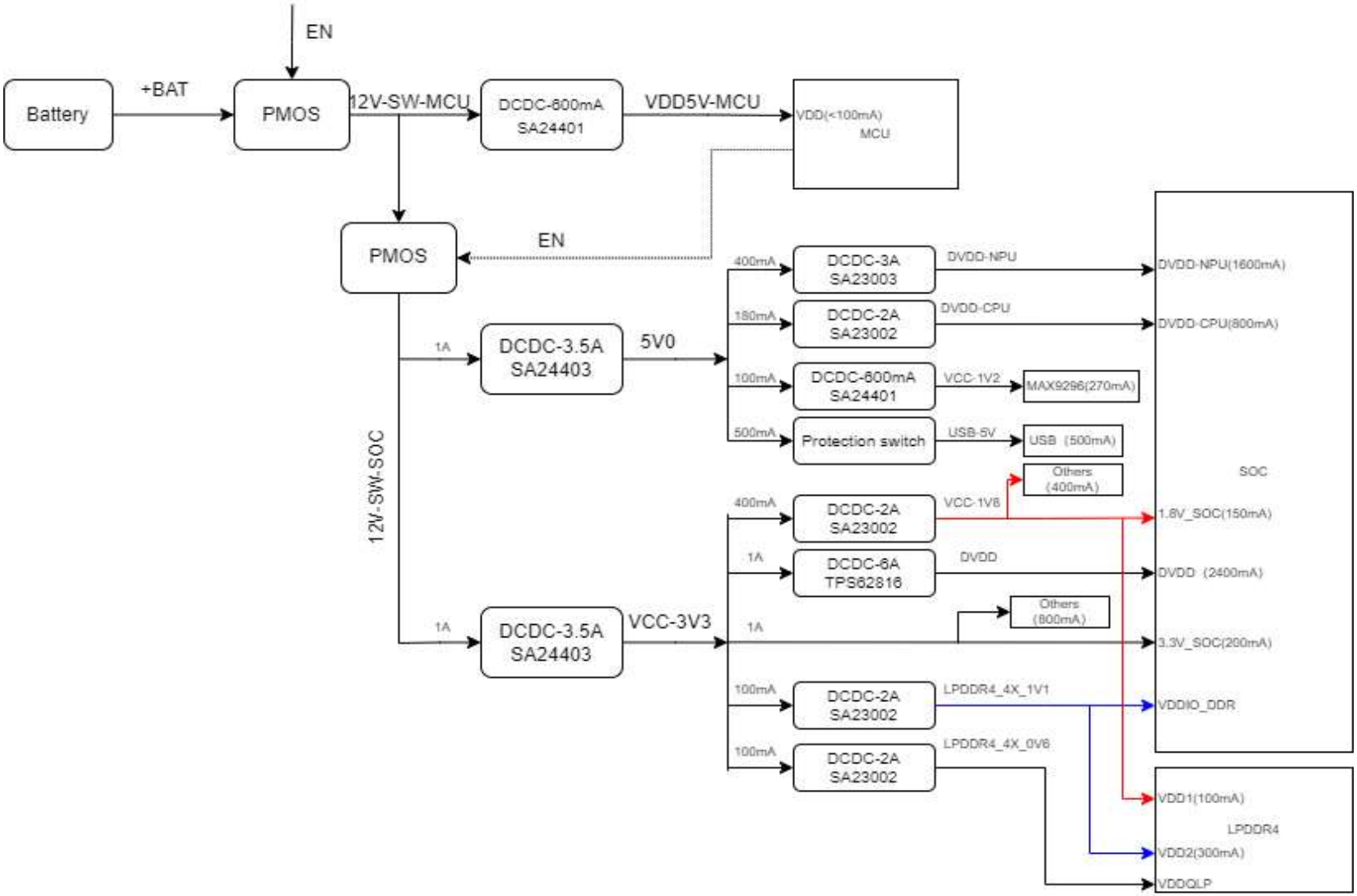
Block diagram

产品方案框图



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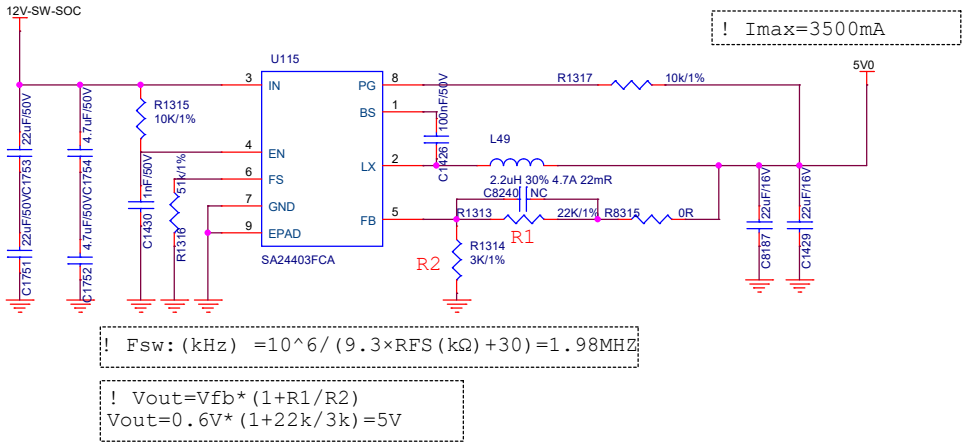
Power Tree



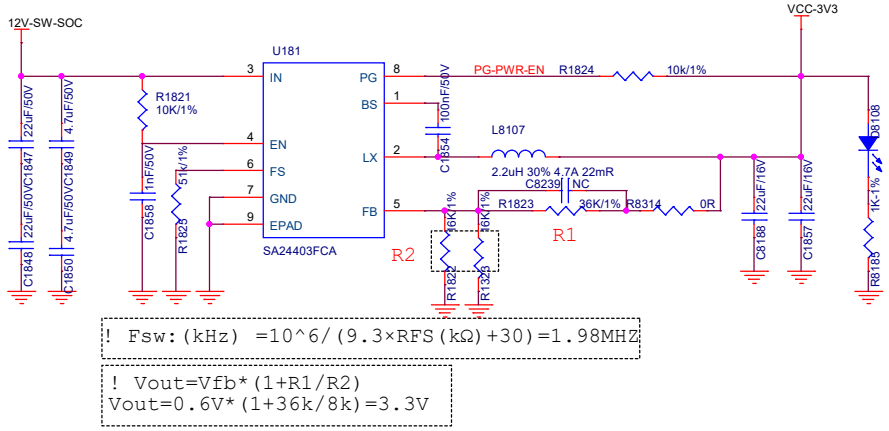
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Power Supply1

12V-->5V

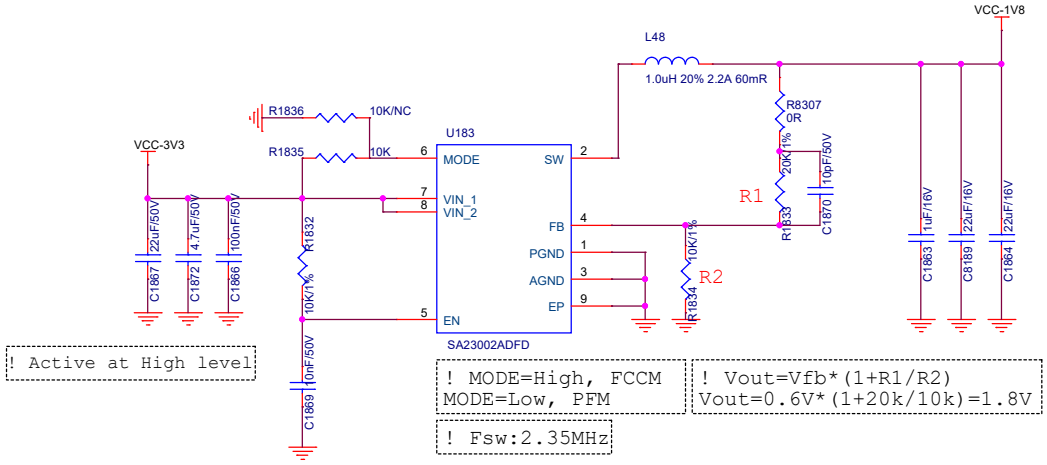


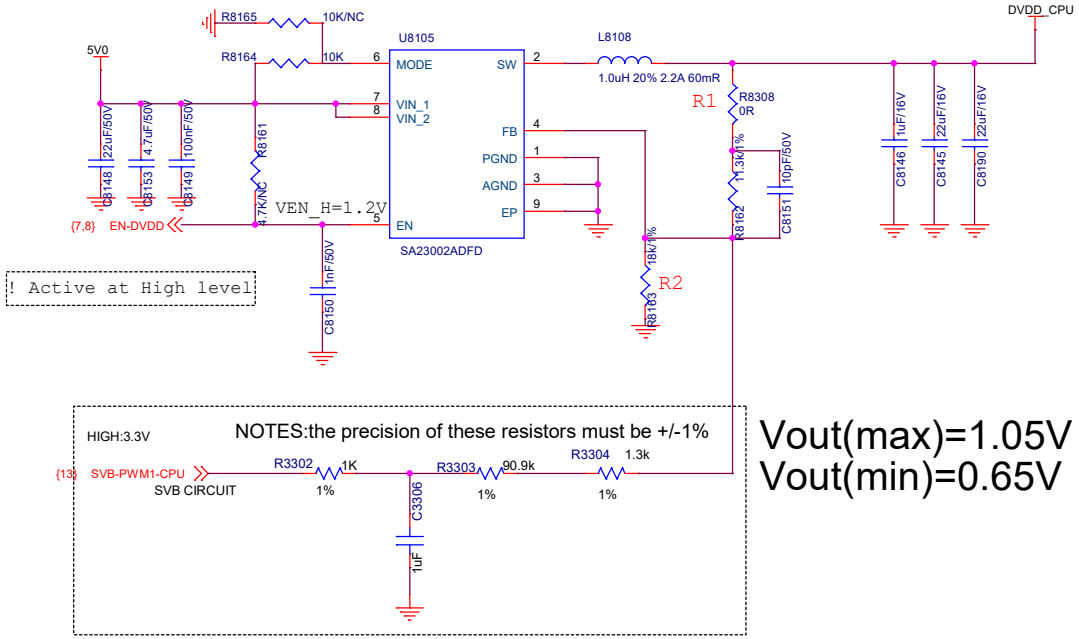
12V-->3V3



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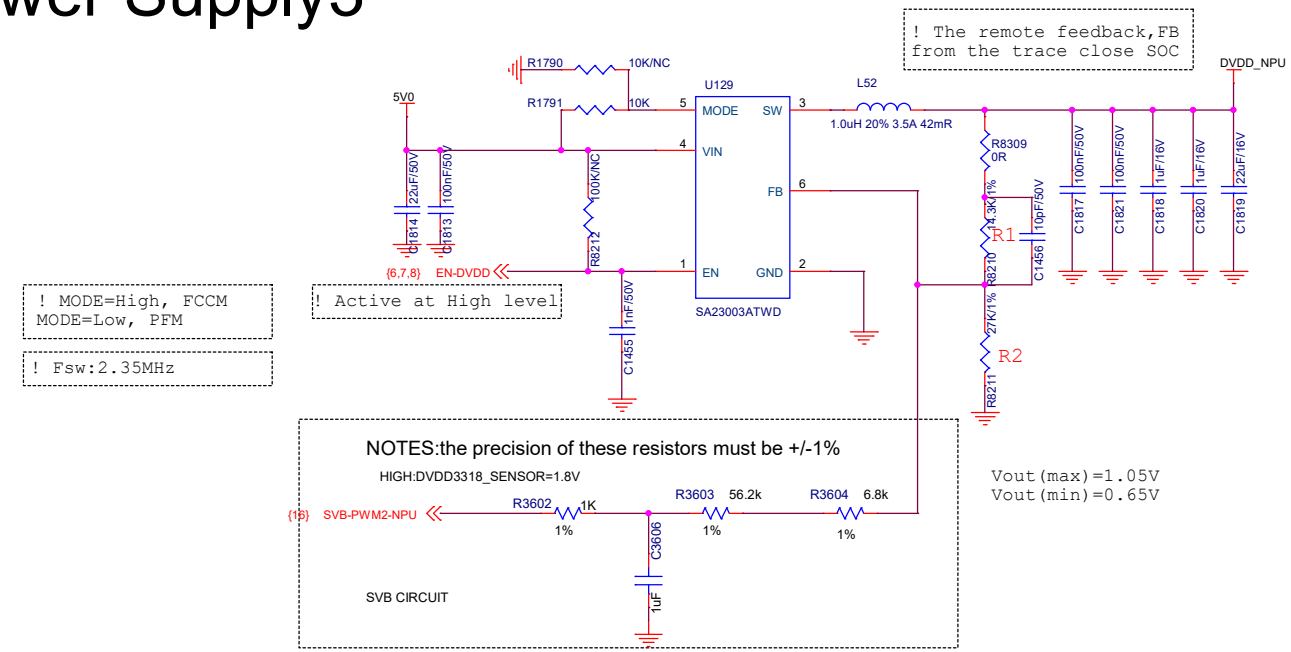
Power Supply3

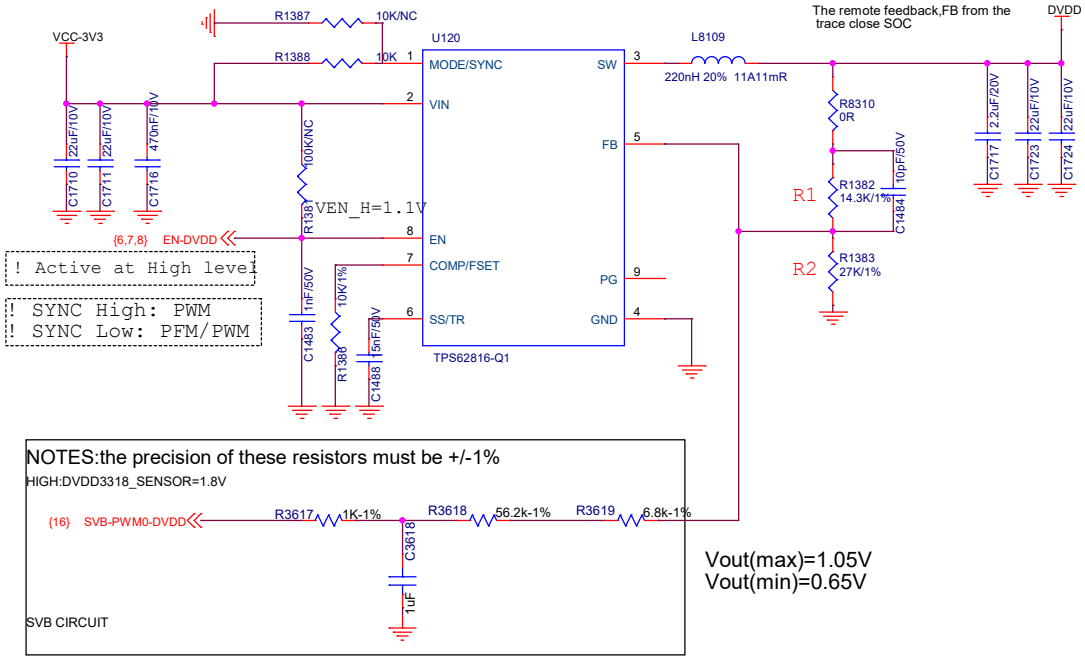




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Power Supply3

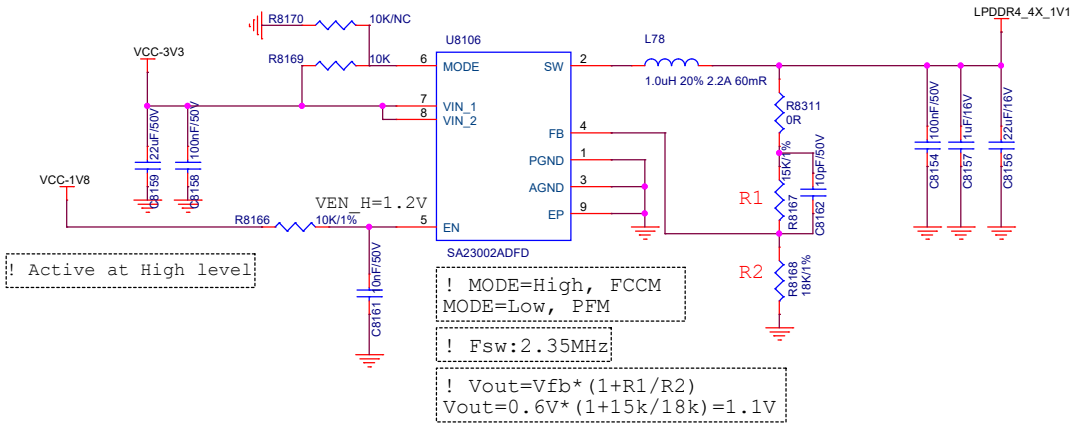


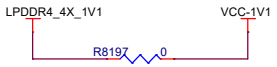


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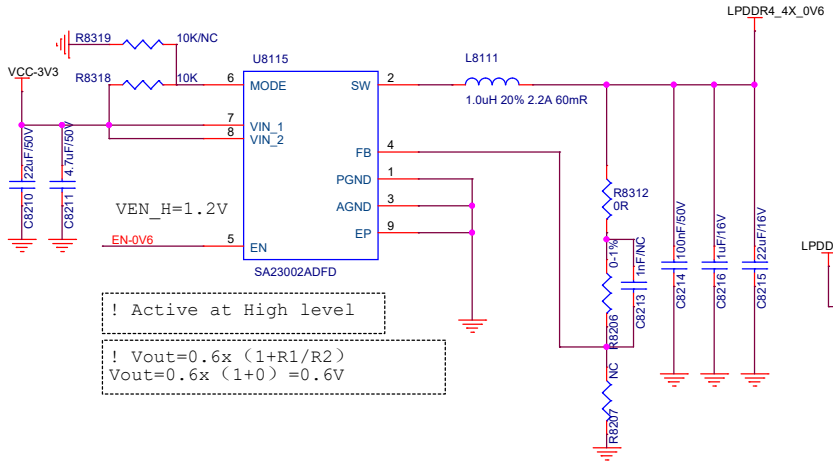
Power Supply4

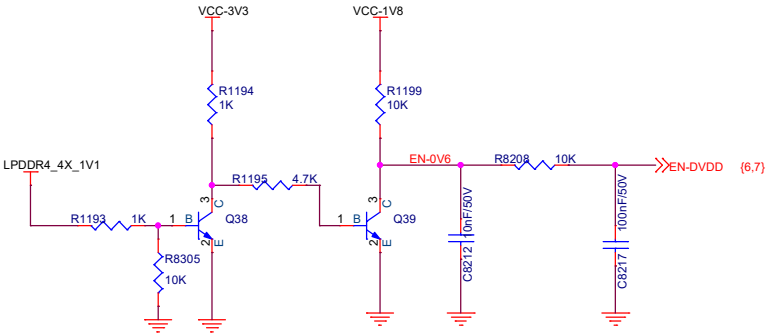
DC/DC 3.3V->1.1V FOR LPDDR4&LPDDR4X





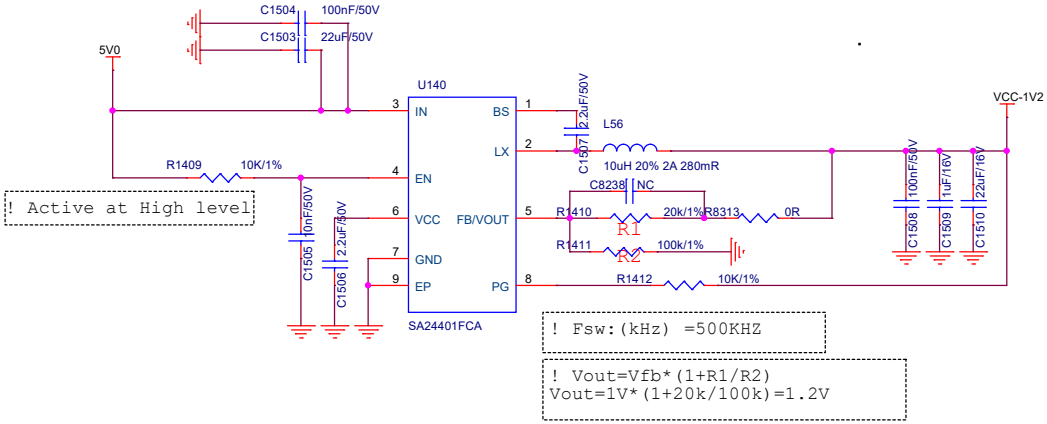
LDO 3.3V->0.6V



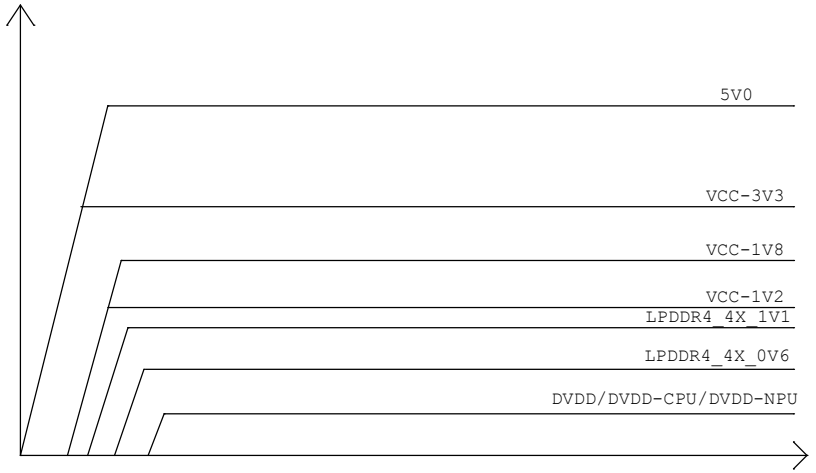


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Power Supply5

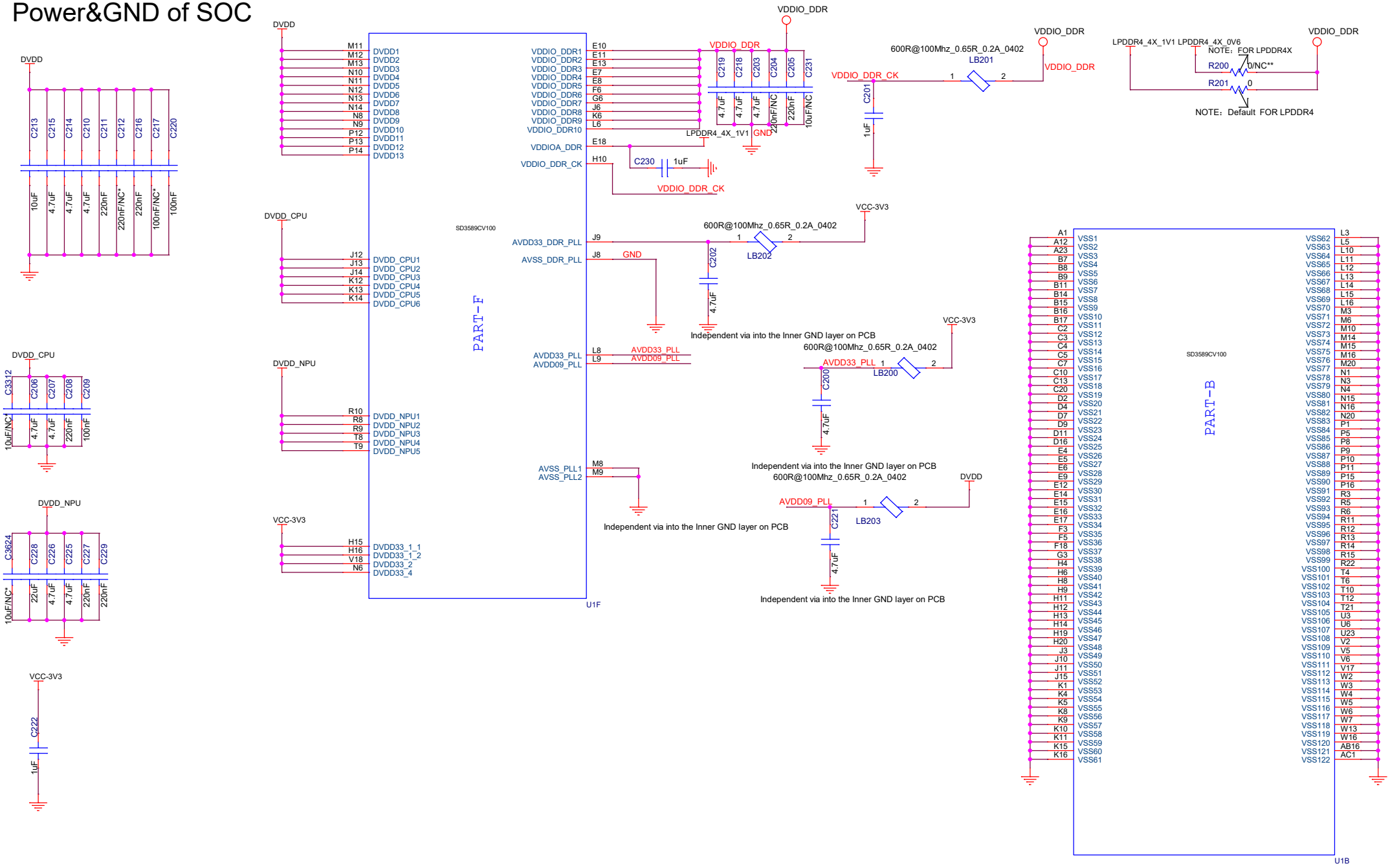


power up sequence



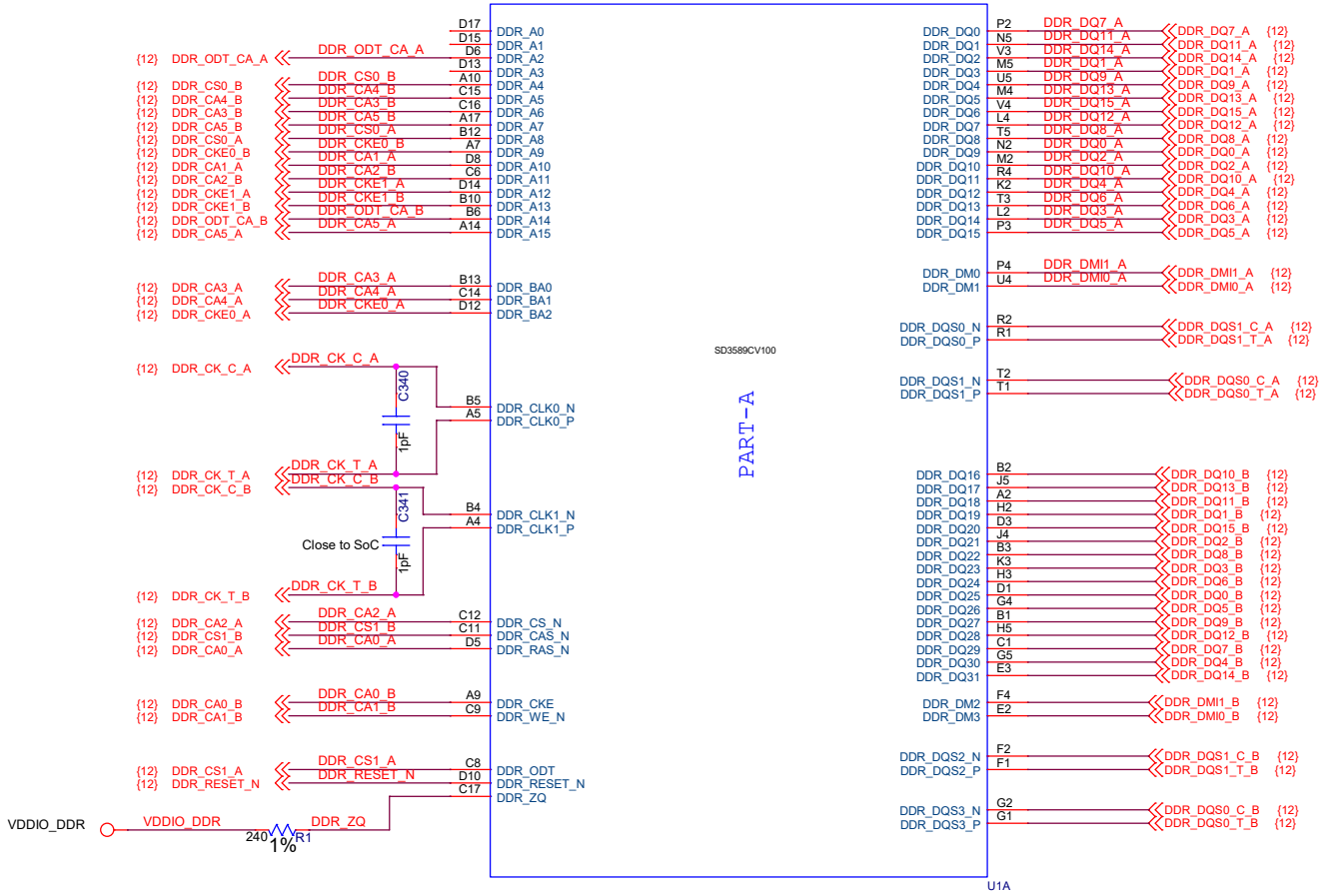
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Power&GND of SOC



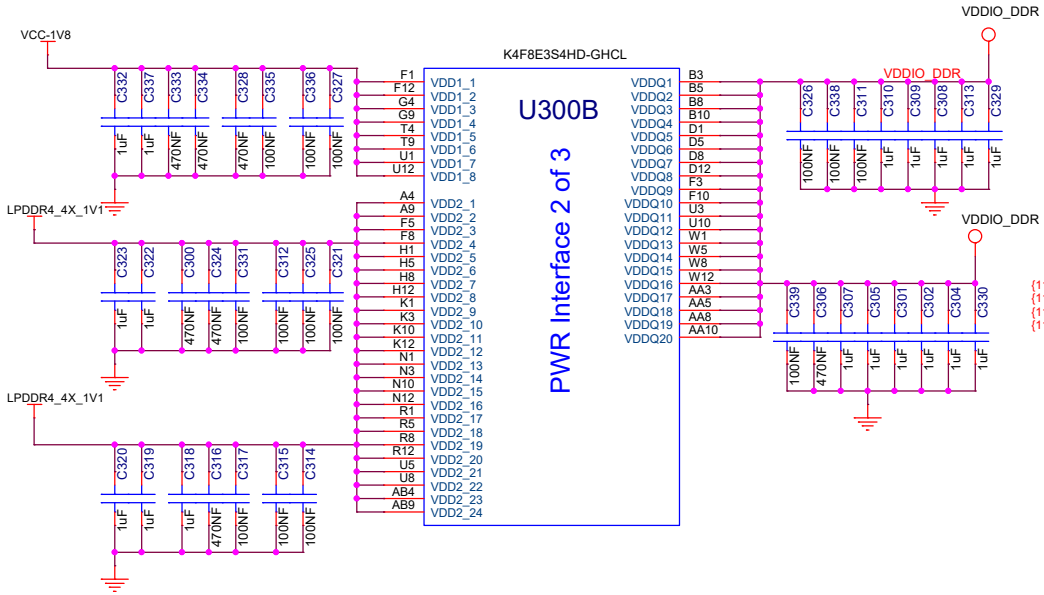
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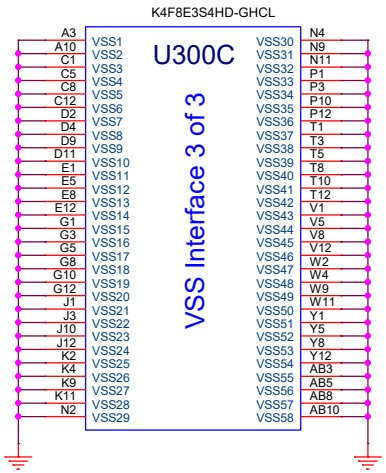
DDRC

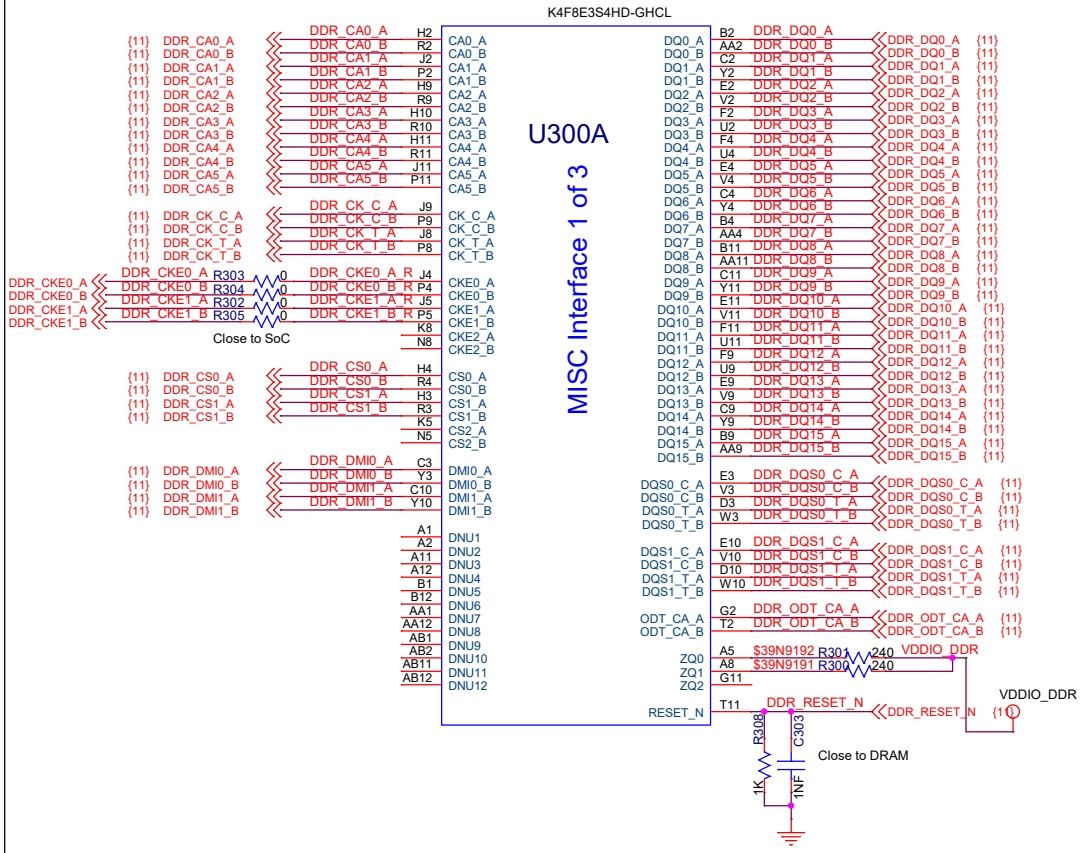


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LPDDR4/4X

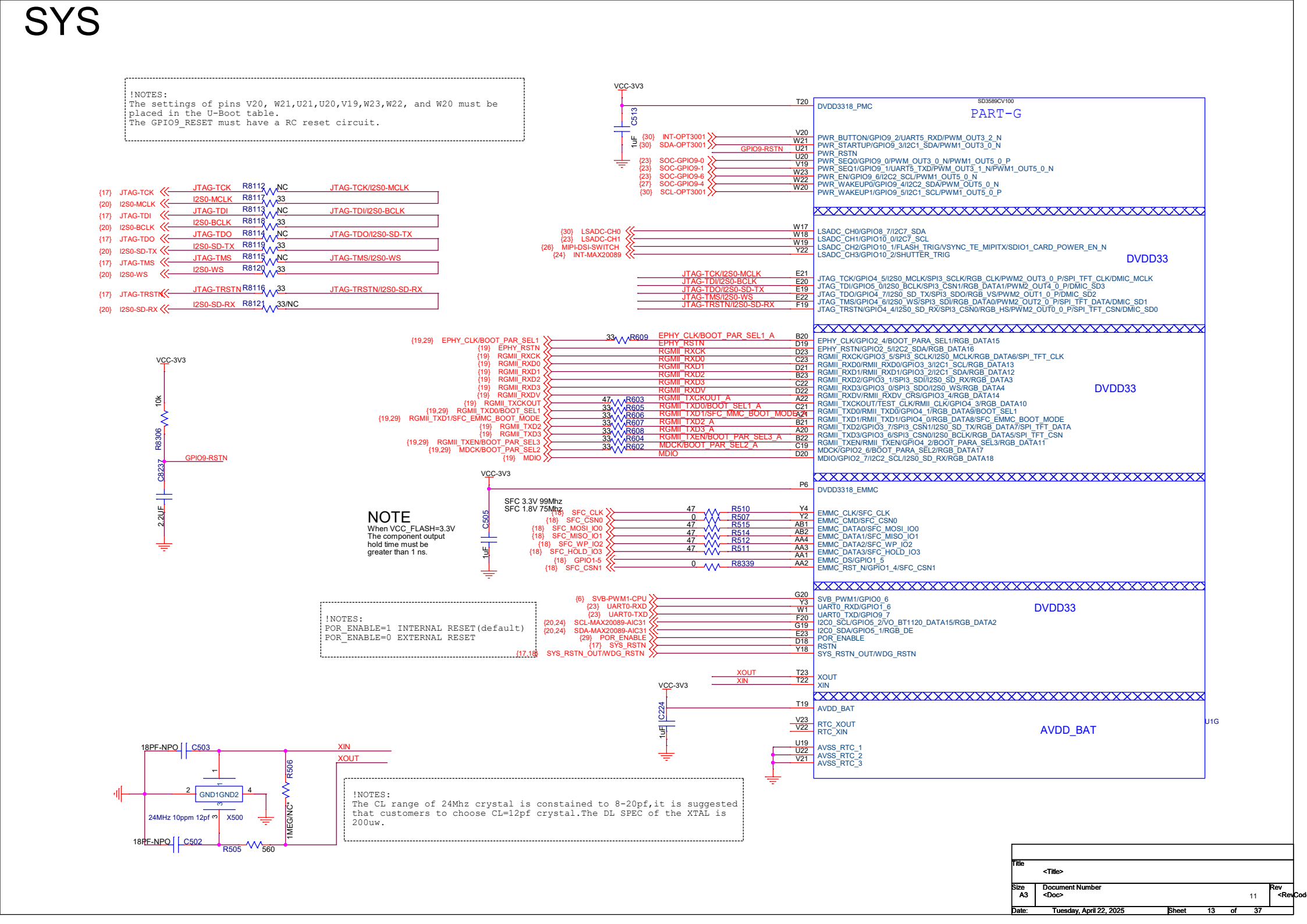






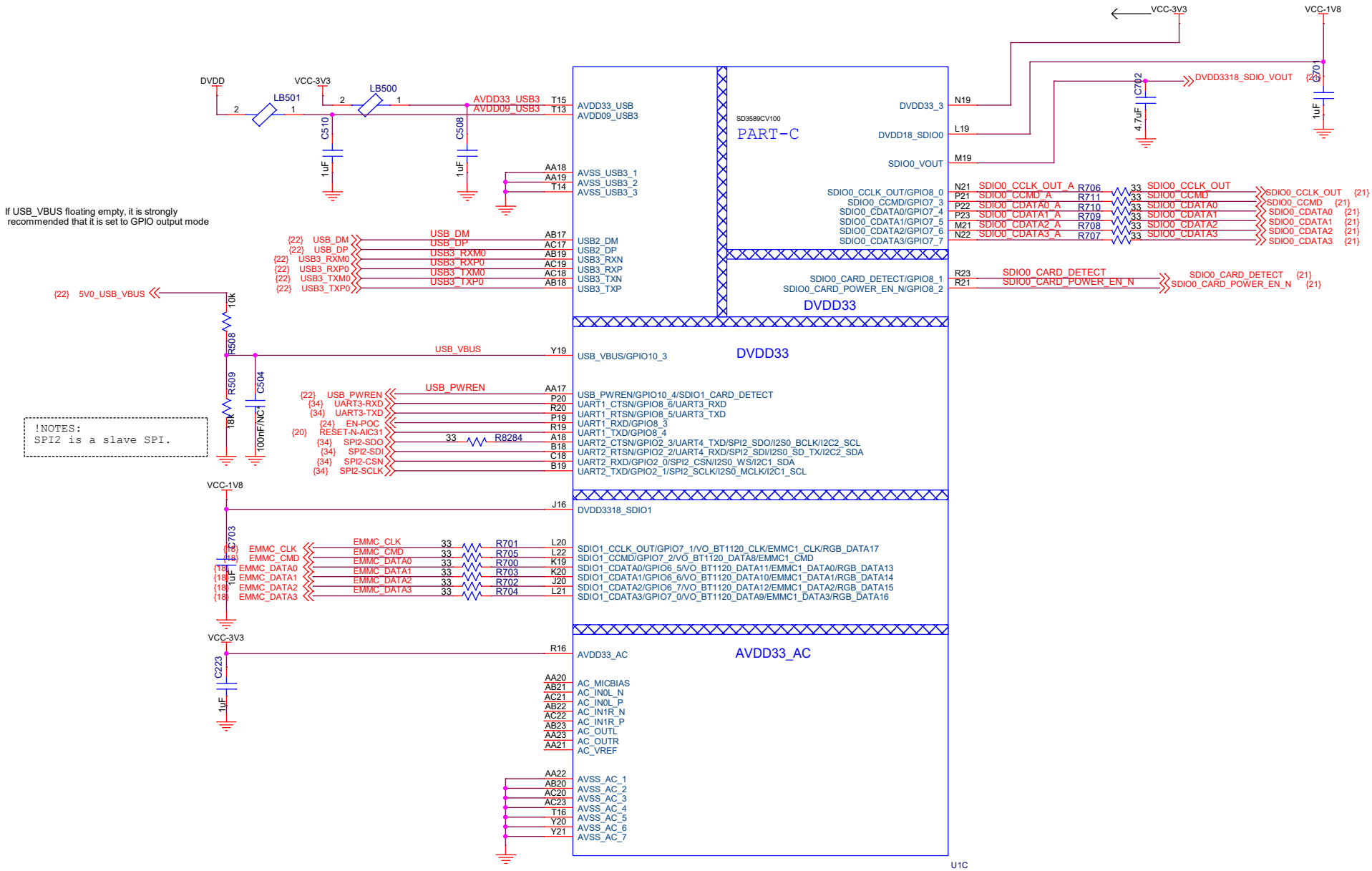
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SYS



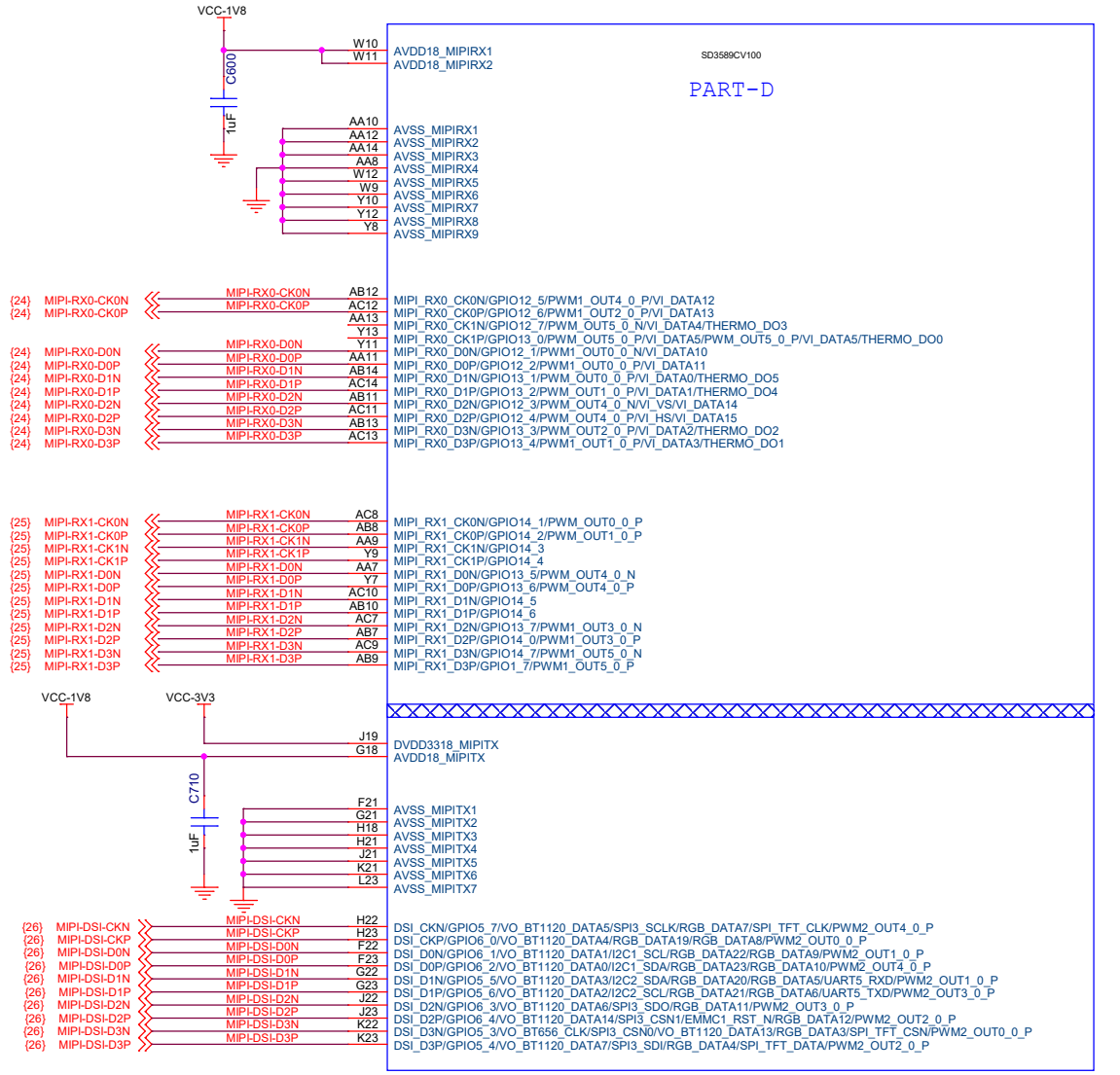
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Peripheral1



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Peripheral1 of SOC



!NOTES:
The MIPI RX differential trace impedance is 100ohm.

!NOTES:
The SD3589 series MIPI RX interface differences

	SD3589CV100	SD3589DV100
MIPI RX interface	MIPI RX0	MIPI RX0
	MIPI RX1	GPIO

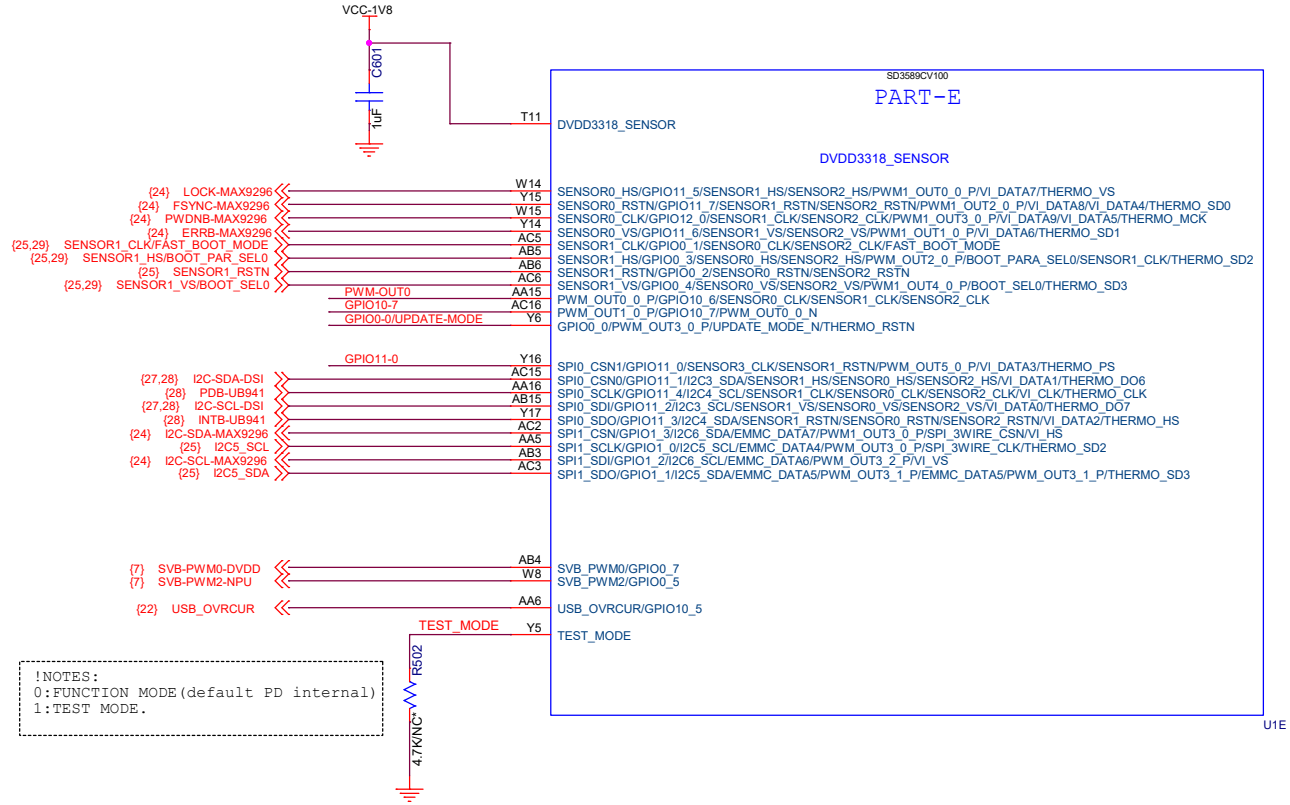
!NOTES:
AVDD18_MIPITX is the MIPI TX analog I/O power.DVDD3318_MIPITX is an I/O power multiplexed with the MIPITX as the non-MIPITX function.

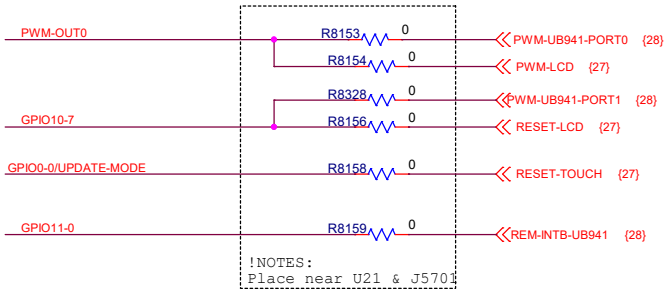
	AVDD18_MIPITX	DVDD3318_MIPITX
MIPITX	1.8V	1.8V or 3.3V
VO_BT1120	1.8V	3.3V
Others	1.8V	3.3V or 1.8V power, depending on the level of the device.

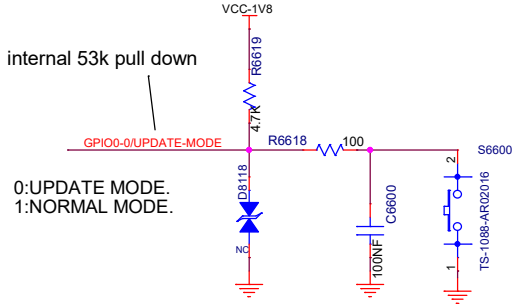
!NOTES:
When the designers use only part of the MIPITX lane,IO multiplexing relation of other lane cannot be used

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Peripheral2 of SOC



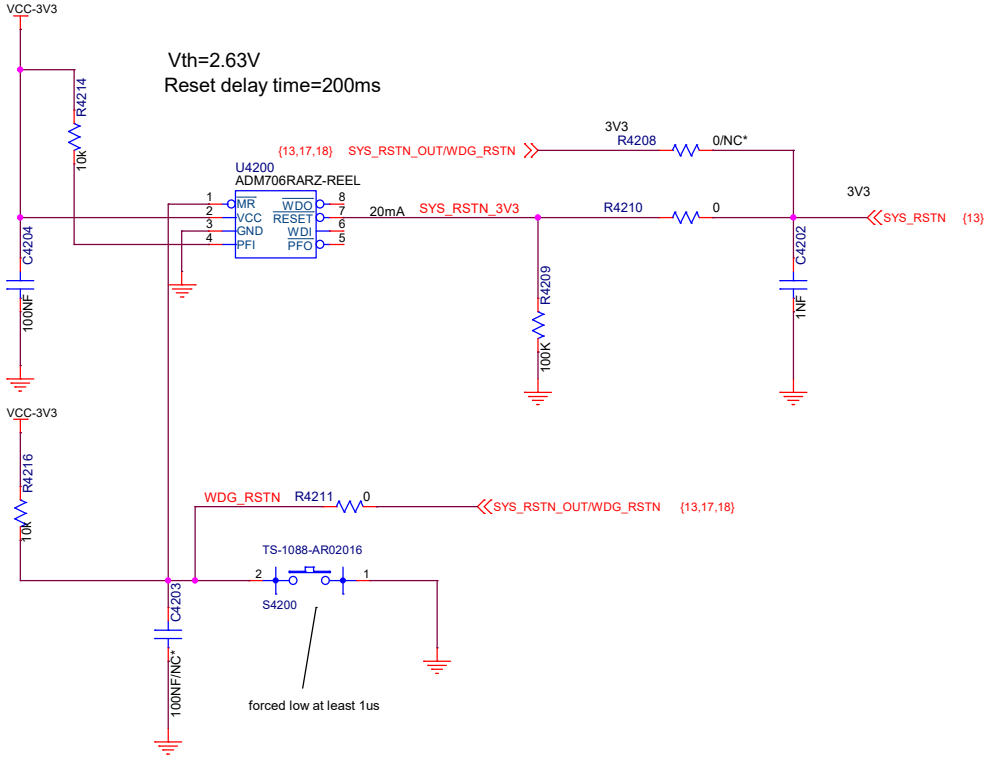


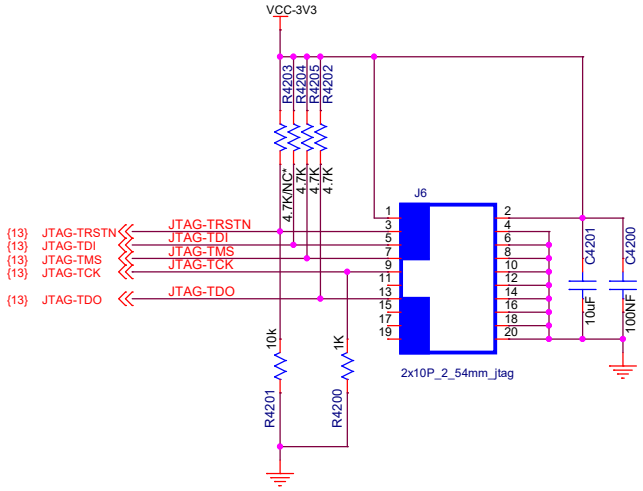


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JTAG

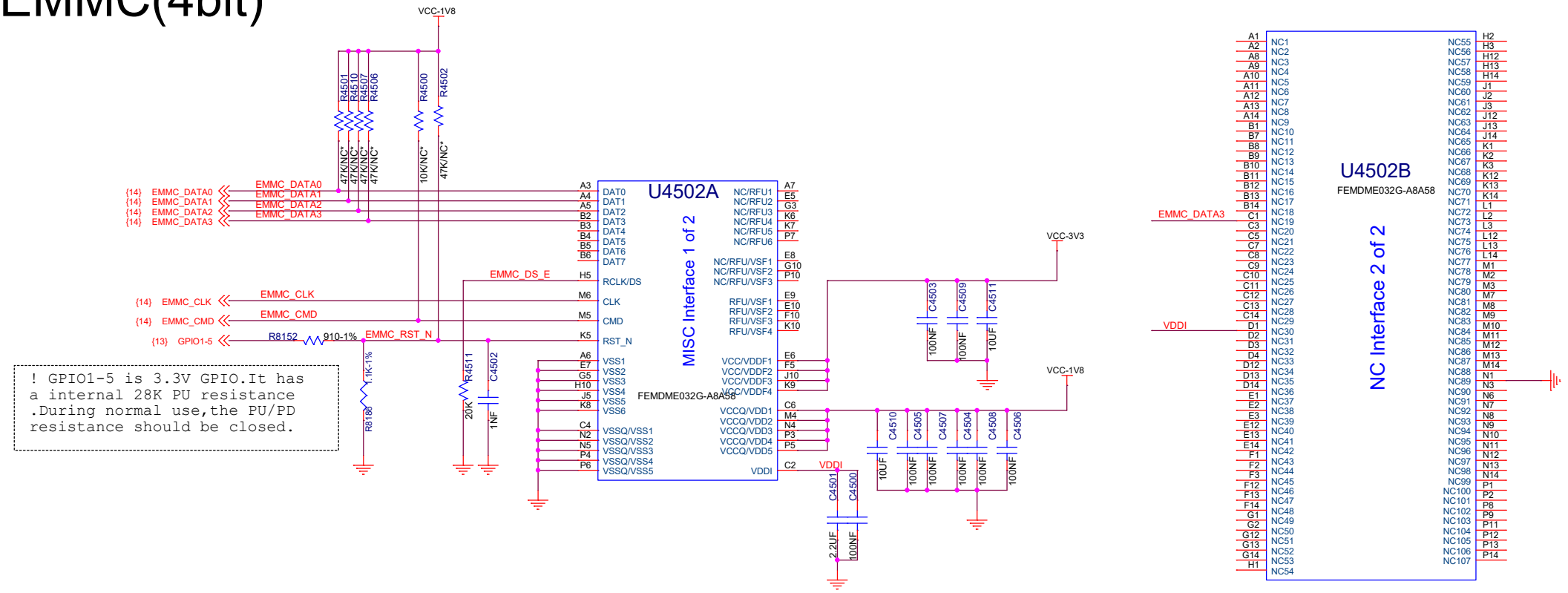
Reset



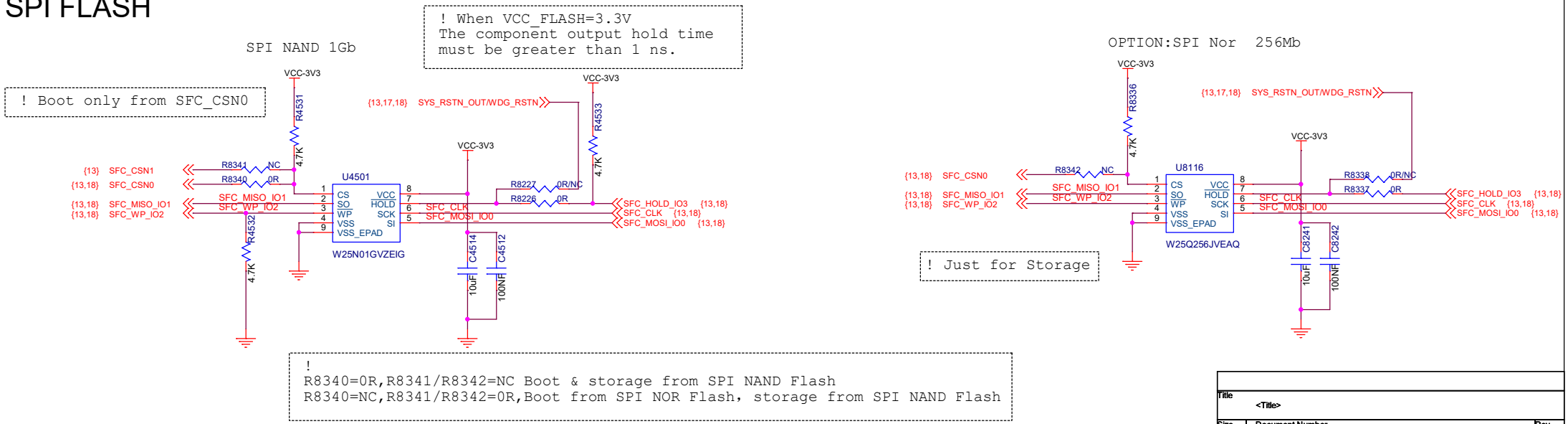


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EMMC(4bit)

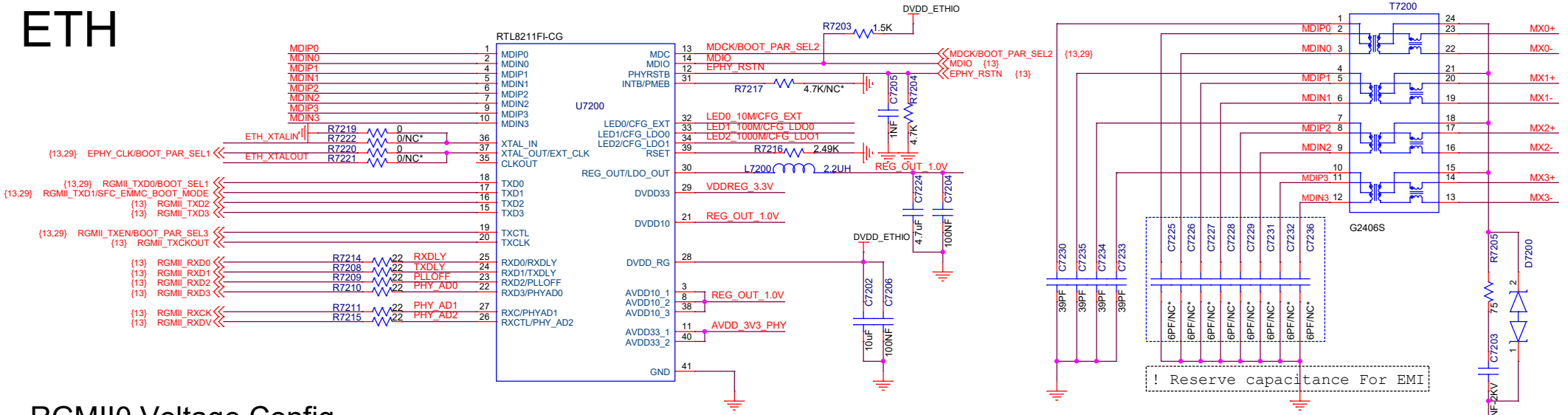


SPI FLASH

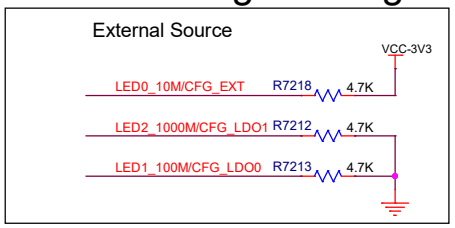


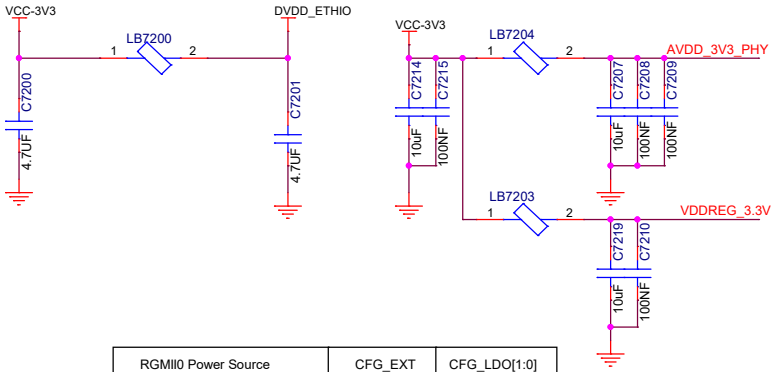
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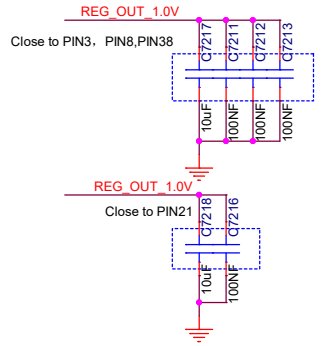
ETH

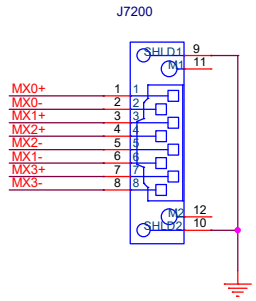


RGMII0 Voltage Config.

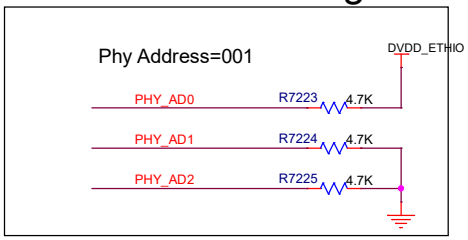








PHY Address Config.



RGMII0 Power Source	CFG_EXT	CFG_LDO[1:0]
External 3.3V(default)	1'b1	2'b00
External 1.8V	1'b1	2'b10
Internal 1.8V	1'b0	2'b10

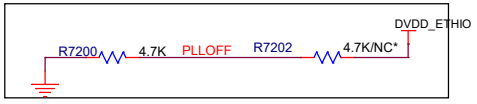
PHY Address	PHYAD[2:0]
1	3'b001 (default)

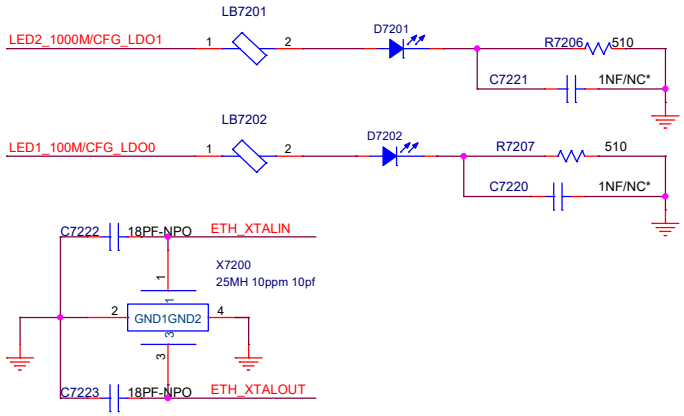
RXDLY=1 Add 2ns delay to RXC for RXD latching
TXDLY=1 Add 2ns delay to TXC for TXD latching
Pull-up disable PLL@ALDPS mode

RGMII0 TXC/RXC Delay Config.



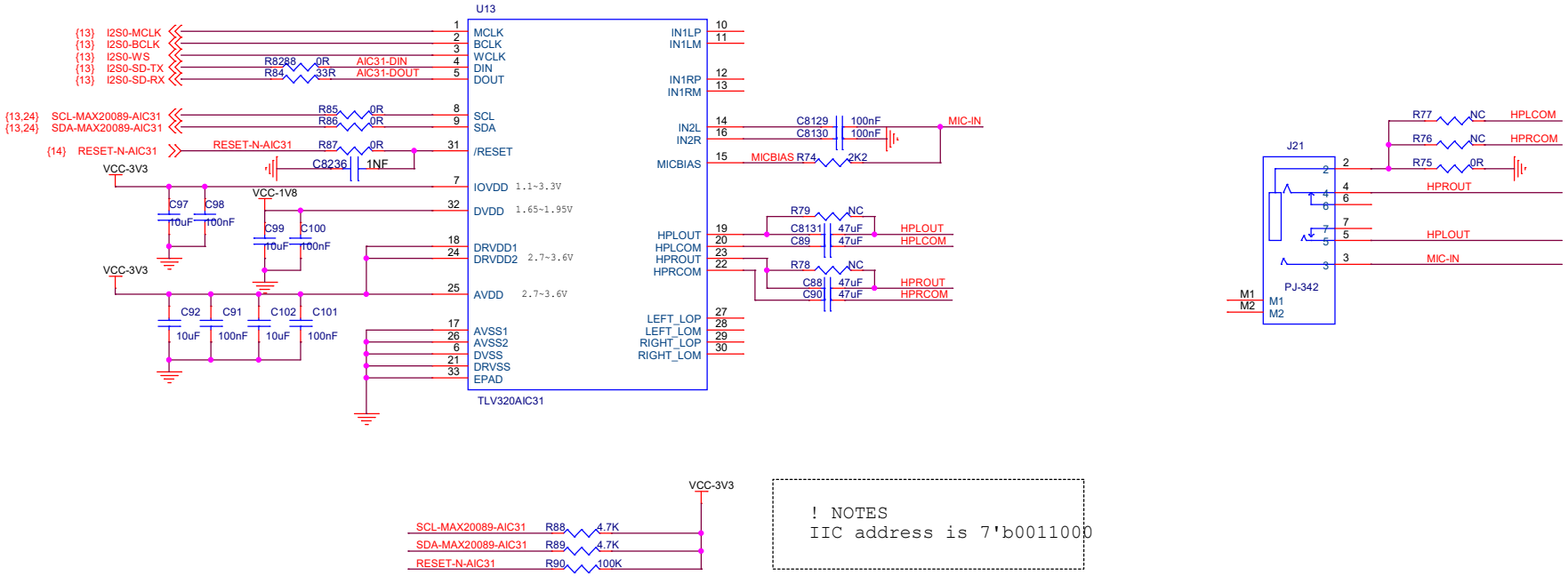
Enable/Disable PLL @ ALDPS





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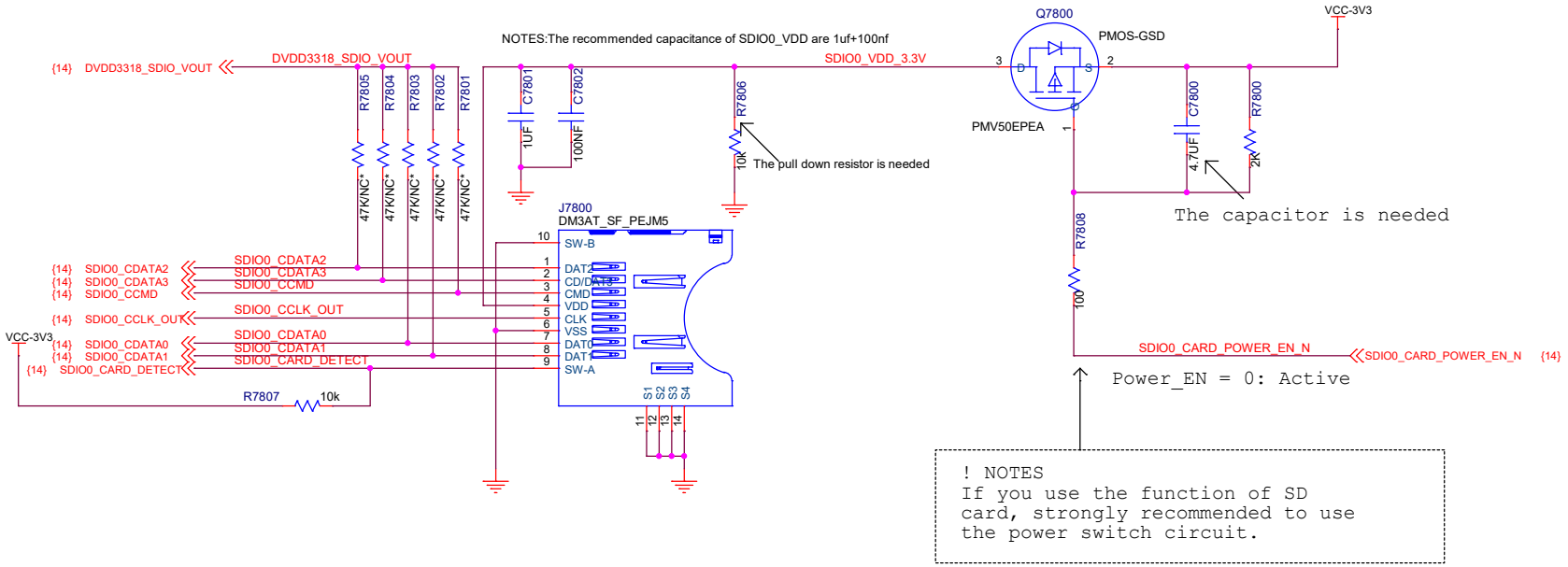
I2S CODEC



! NOTES
IIC address is 7'b0011000

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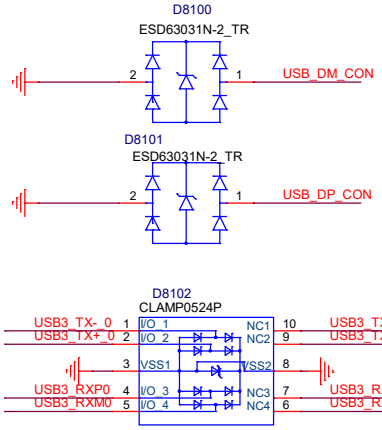
Micro-SD Card



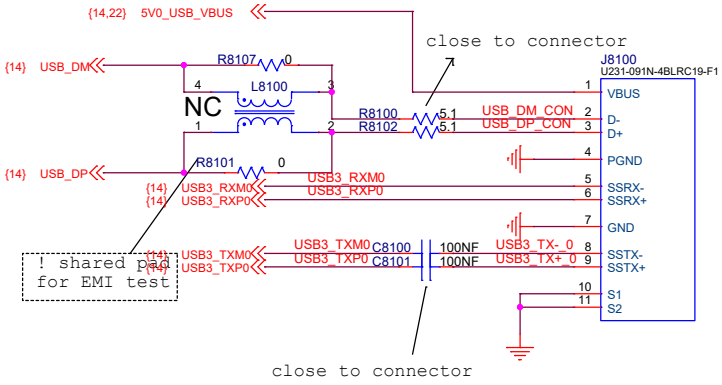
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USB

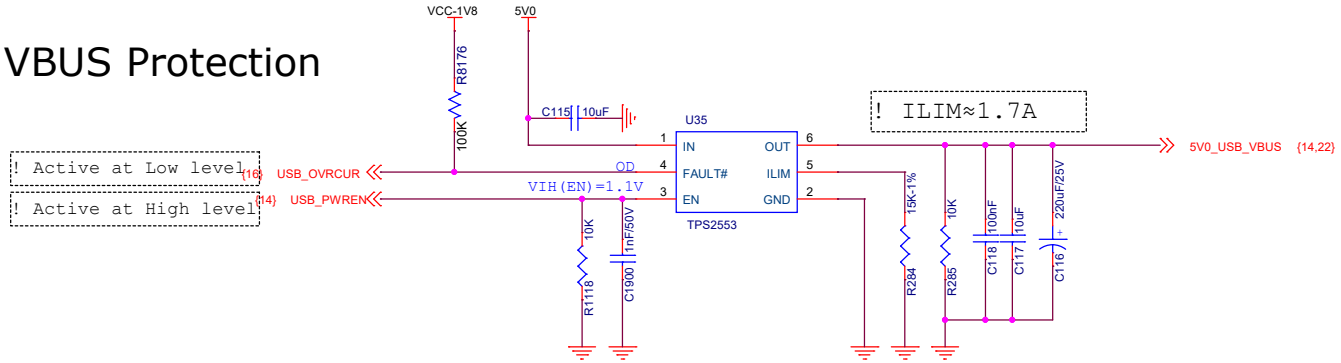
USB2.0:It is recommended to select a TVS with Ipp>=4A,Cj<=1.2pF,Vbr<=14V, and without secondary snap back



USB3.0:It is recommended to select a TVS with Ipp>=4A,Cj<=0.65pF,Vbr<=7V, vclamp<=10V(@Ipp=4A).

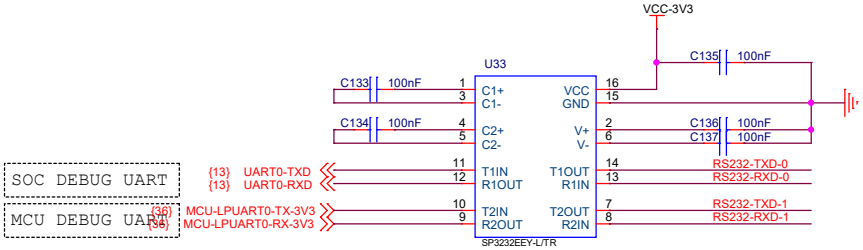


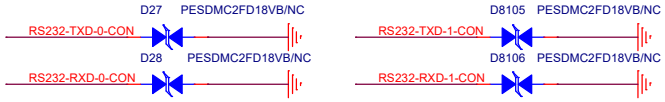
VBUS Protection



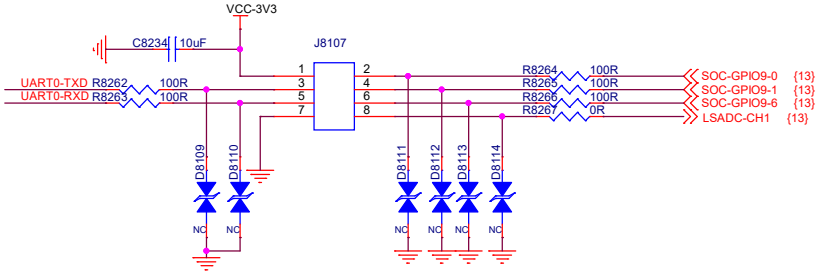
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UART0(DEBUG)

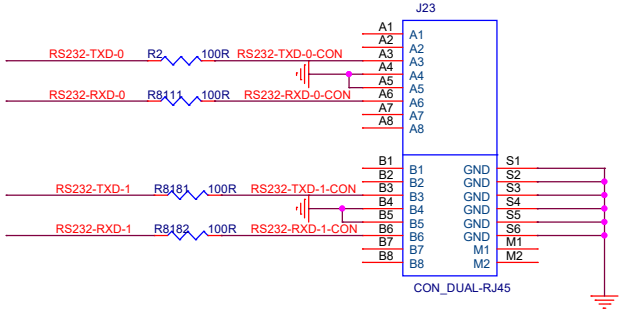




TTL UART & GPIO



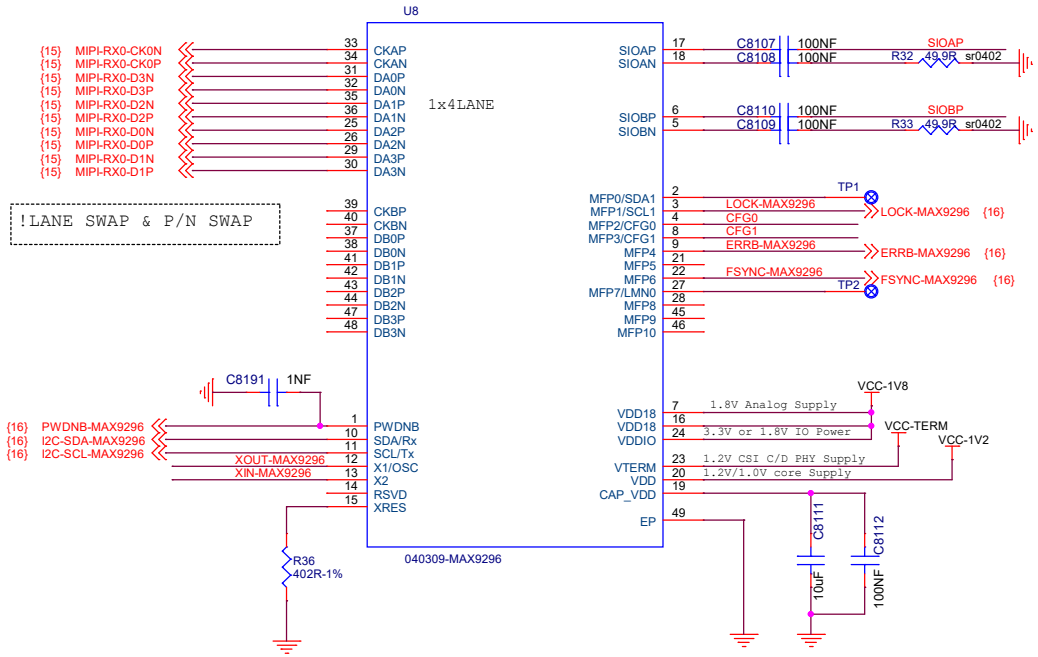
!SICKSCREEN:
UPSIDE:SOC UART
DOWNSIDE:MCU UART

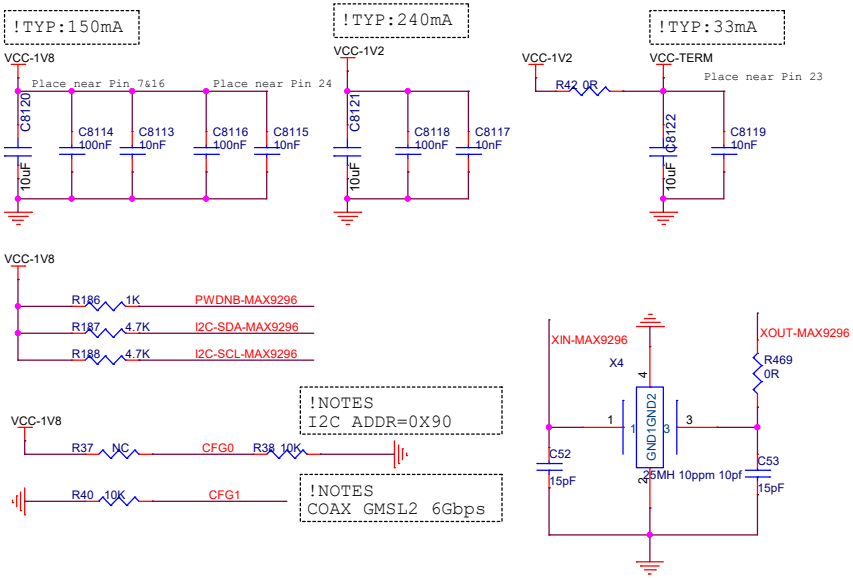


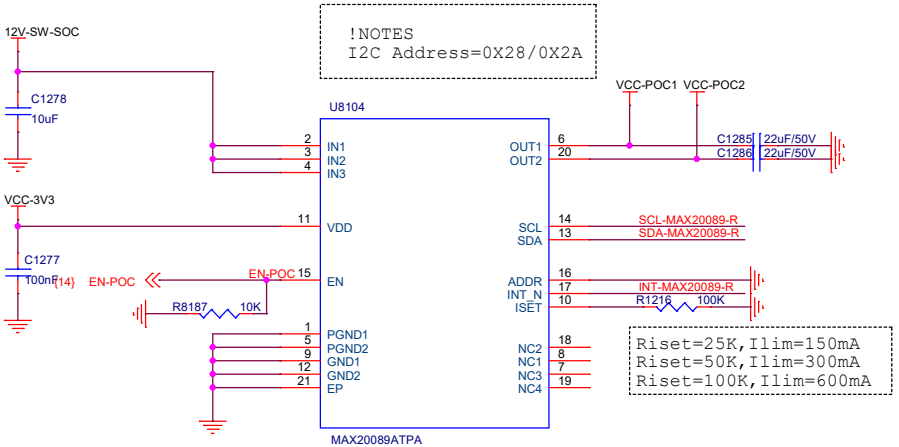
Cisco,HUAWEI console lane
RJ45 to DB9(FEMALE)

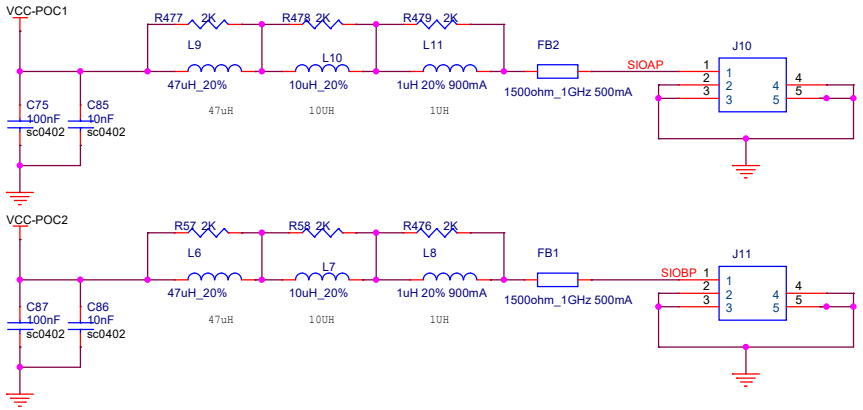
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DeSerializer





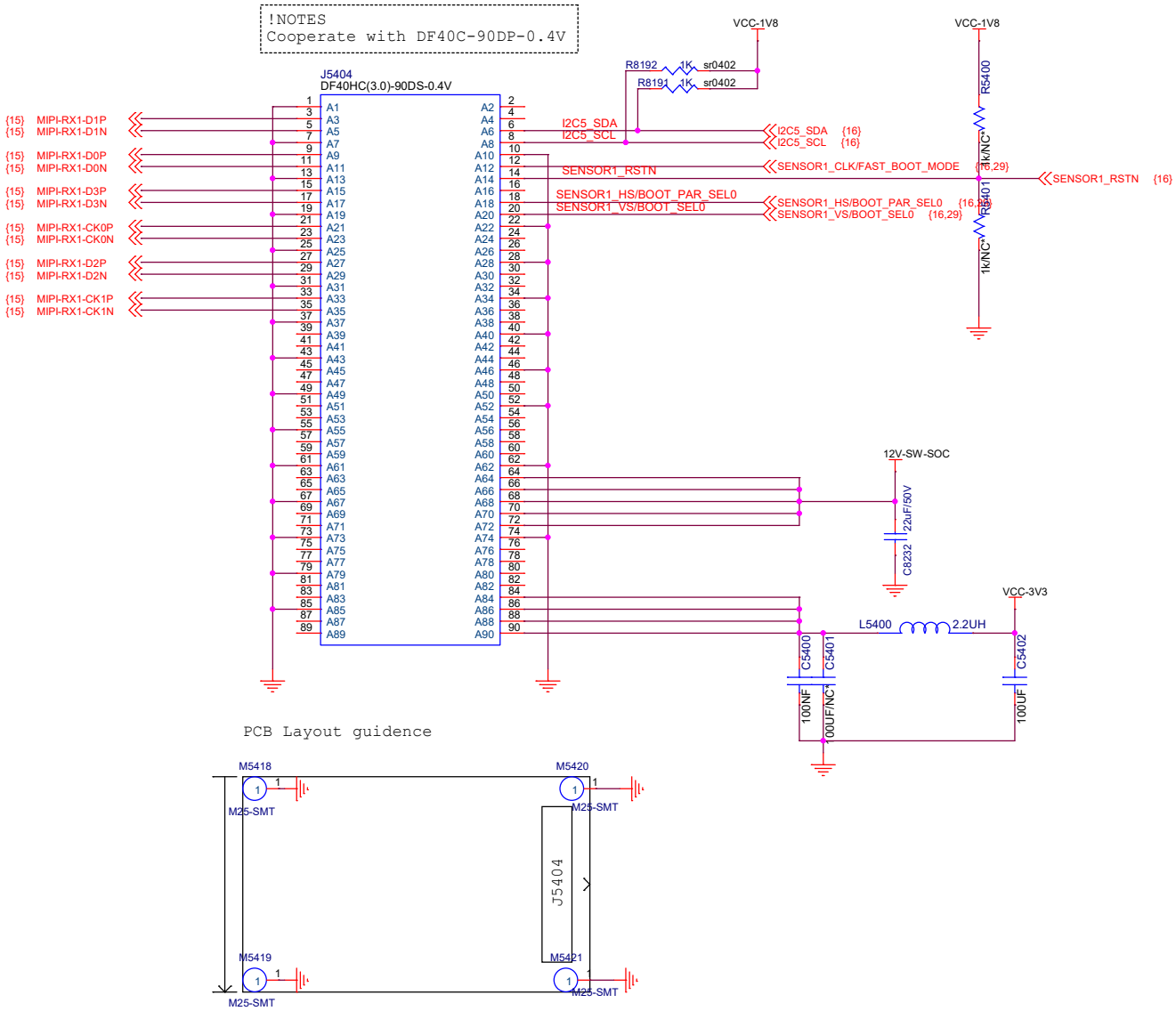






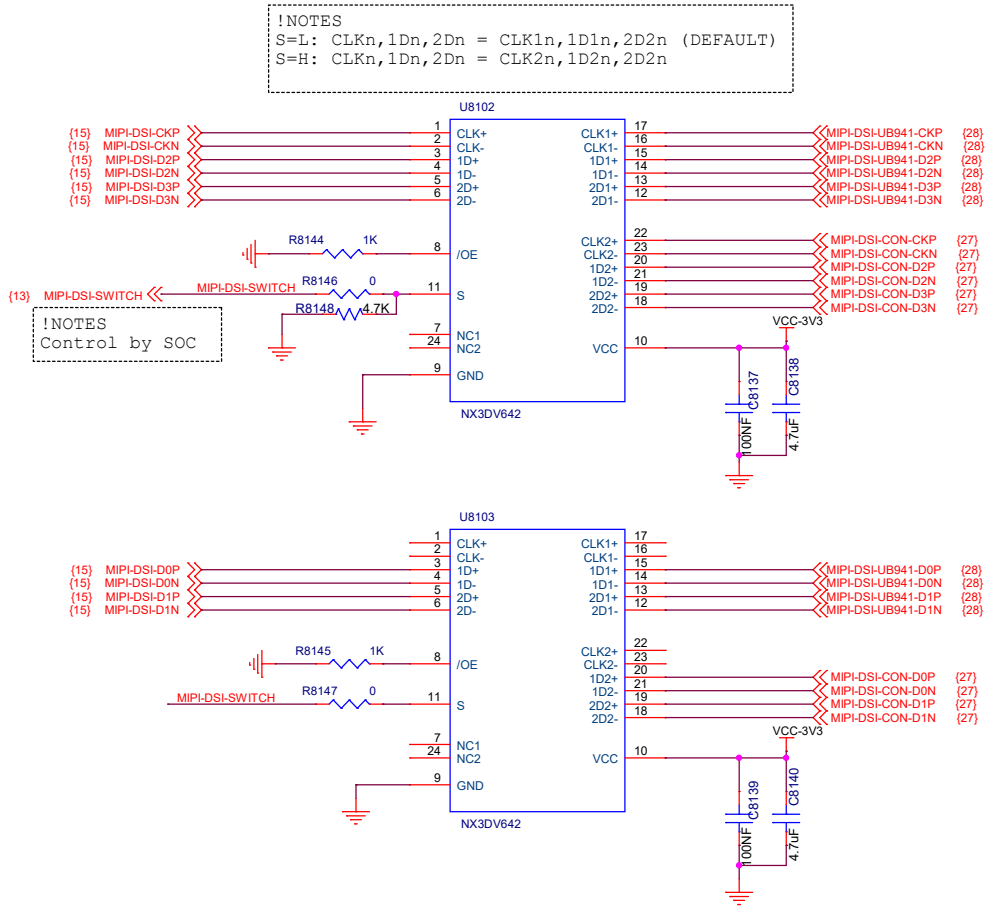
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Sensor Connector



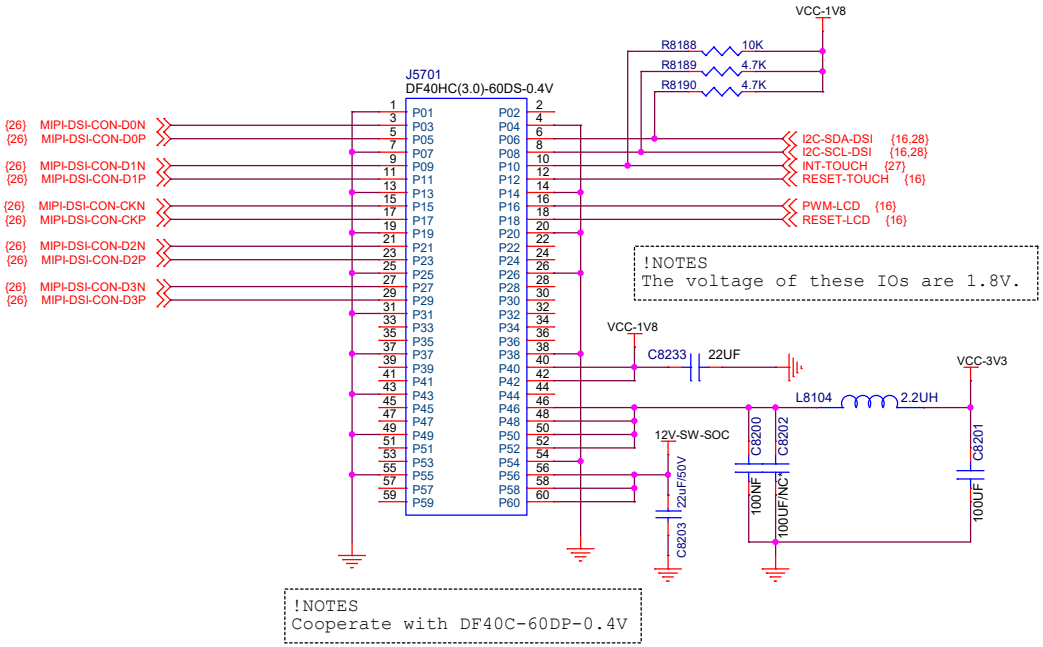
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Size A3	Document Number <Doc>		Rev <RevCode> 24
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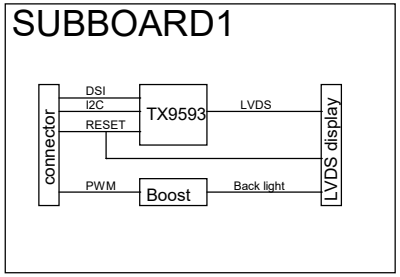
MIPI DSI SWITCH(JUT FOR DEBUG)

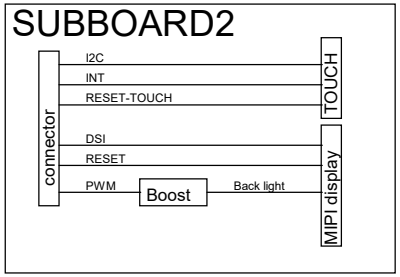


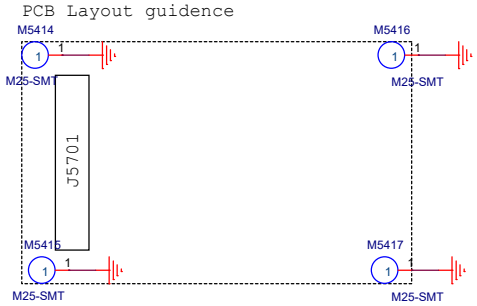
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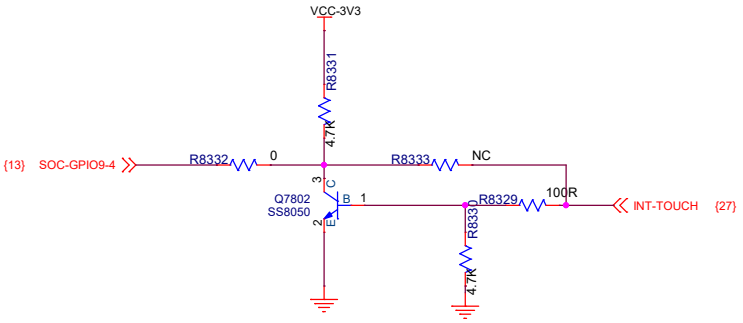
MIPI DSI CONNECTOR











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MIPI DSI Serializer

! Imax=150mA

! Imax=30mA

! P/N SWAP & LANE SWAP

! Differential impedance 100ohm±10

! NOTE
1,Port 0 use GPIO1 to control backlight of remote LED;
2,Port 0 use D_GPIO1 to control backlight of remote LED;
3,Port 0 use REM_INTB to control the INT signal of touch screen;
3,Port 1 use INTB to control the INT signal of touch screen;

! Active at Low level

! Active at Low level

! PDB = 1, device is enabled (normal operation)
PDB = 0, device is powered down.

! Share pads

! Differential impedance 100ohm±10

! I2C address:0x0C

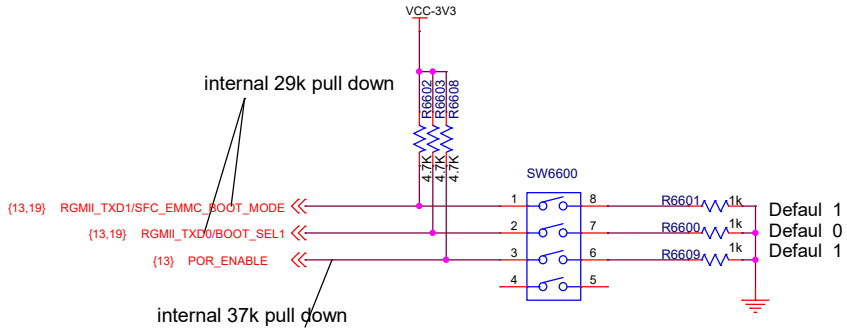
! INTB = H, Normal Operation
INTB = L, Interrupt Request

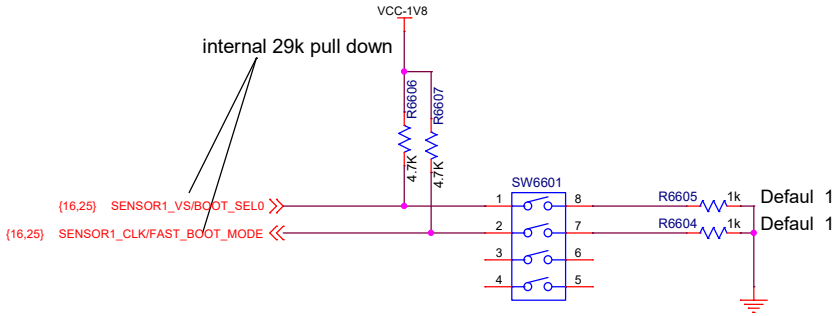
PCB Silkscreen:
"Vedio out 0"

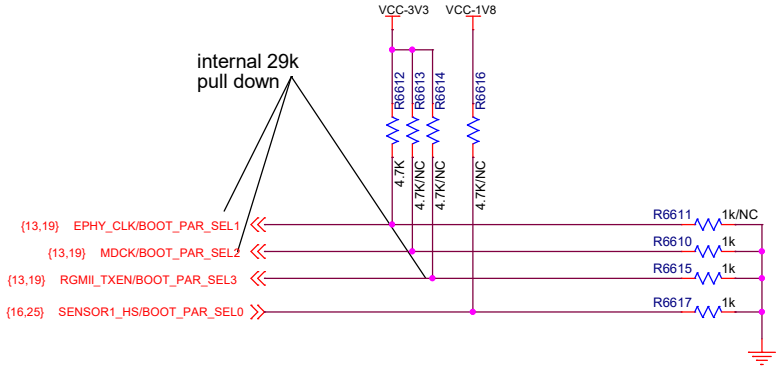
PCB Silkscreen:
"Vedio out 1"

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Hardware Configuration







BOOT_SEL[1:0]	SFC_EMMC_BOOT_MODE	MODE
00	0	SPI Nor Flash 3Byte
00	1	SPI Nor Flash 4Byte
01	0	SPI Nand Flash 1 wire
01	1	SPI Nand Flash 4 wire
11	0	EMMC 4bit
11	1	EMMC 8bit

FAST_BOOT_MODE	MODE
0	Normal BOOT
1	Fast BOOT

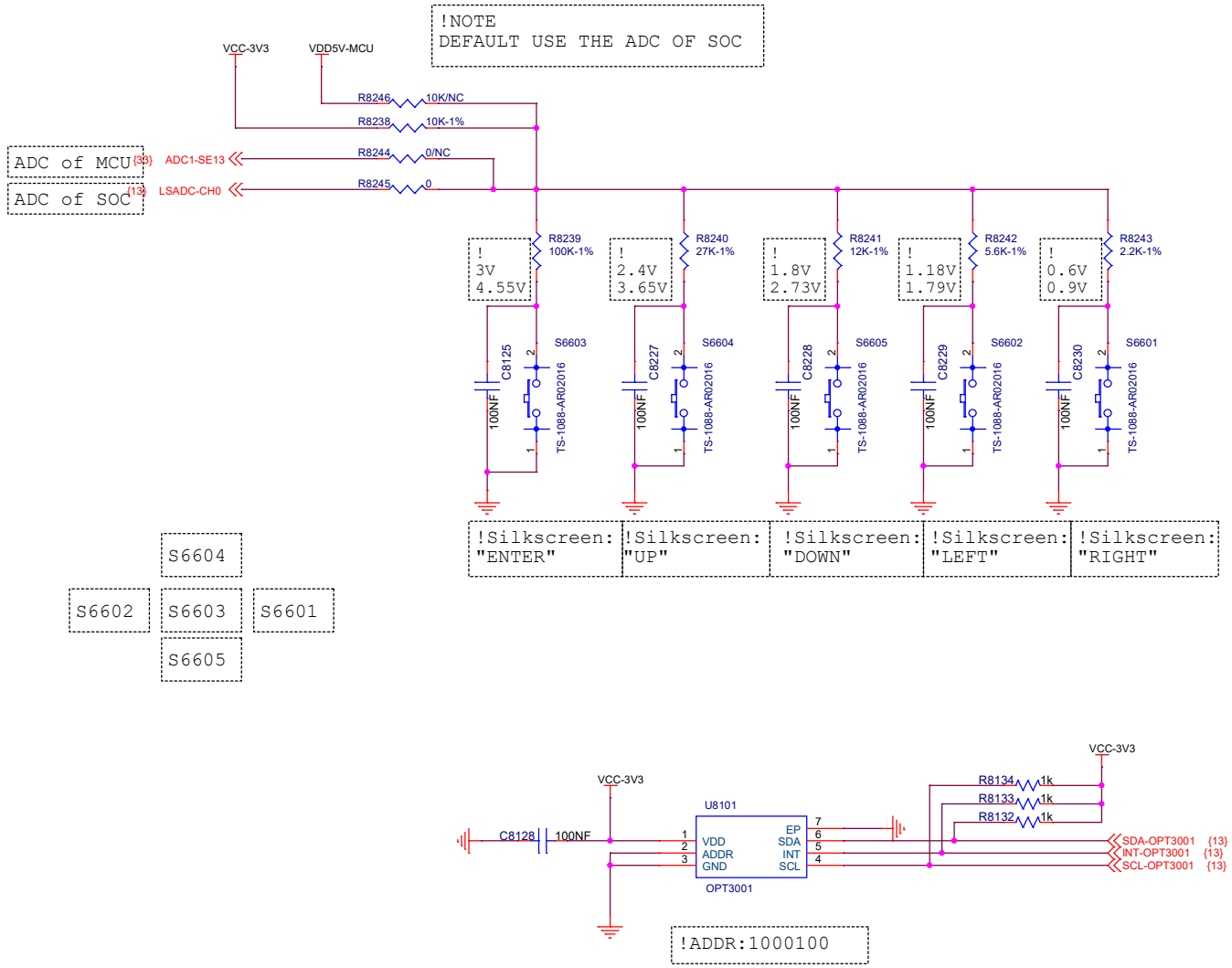
POR_ENABLE	Function
0	disable
1	enable

BOOT_PAR_SEL[2:1:0]	Type	Speed	Capacity	Topology	PCB layer
010	LPDDR4	TBD	1GB	T	6

BOOT_PAR_SEL3	Function
0	SFC 3.3V 99Mhz
1	SFC 1.8V 75Mhz

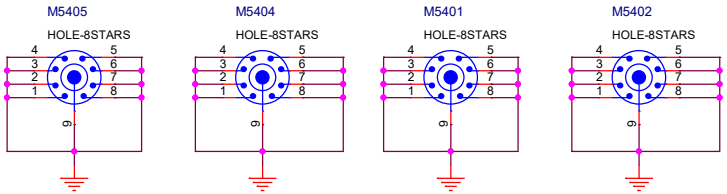
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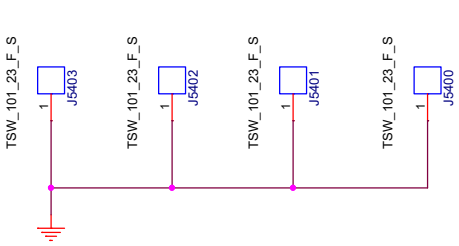
BUTTON & Light Sensor

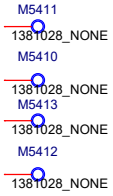


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BUTTON







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MCU Domain

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MCU

!NOTES
LPUART0 is used for debug.

!NOTES
LSPI1 use for communication with SOC,and it is a master SPI
LPUART2 use for communication with SOC.

U8108

{36} ADC0-SE0 — ADC0-SE0 — 79 — PTA0/FTM2_CH1/LPI2C0_SCLS/FXIO_D2/FTM2_QD_PHA/LPUART0_CTS/TRGMUX_OUT3/ADC0_SE0/CMP0_IN0
{36} ADC0-SE1 — ADC0-SE1 — 78 — PTA1/FTM1_CH1/LPI2C0_SDAS/FXIO_D3/FTM1_QD_PHA/LPUART0_RTS/TRGMUX_OUT0/ADC0_SE1/CMP0_IN1
{36} MCU-LPUART0-RX — MCU-LPUART0-RX — 73 — PTA2/FTM3_CH0/LPI2C0_SDA/EWM_OUT_b/FXIO_D4/LPUART0_RX/ADC1_SE0
{36} MCU-LPUART0-TX — MCU-LPUART0-TX — 72 — PTA3/FTM3_CH1/LPI2C0_SCL/EWM_IN/FXIO_D5/LPUART0_TX/ADC1_SE1
{36} SWD-DIO — SWD-DIO — 98 — PTA4/CMP0_OUT/EWM_OUT_b/JTAG_TMS/SWD_DIO
MCU-RESET — 97 — PTA5/TCLK1/RESET_b
{36} ADC0-SE2 — ADC0-SE2 — 58 — PTA6/FTM0_FLT1/LPSPI1_PCS1/LPUART1_CTS/ADC0_SE2
{36} ADC0-SE3 — ADC0-SE3 — 57 — PTA7/FTM0_FLT2/RTC_CLKIN/LPUART1_RTS/ADC0_SE3
{36} MCU-DEBUG-LED1 — MCU-DEBUG-LED1 — 100 — PTA8/LPUART2_RX/LPSPI2_SOUT/FXIO_D6/FTM3_FLT3
{36} MCU-DEBUG-LED2 — MCU-DEBUG-LED2 — 99 — PTA9/LPUART2_TX/LPSPI2_PCS0/FXIO_D7/FTM3_FLT2/FTM1_FLT3
{35} MCU-PTA10 — 92 — PTA10/FTM1_CH4/FXIO_D0/JTAG_TDO/noetm_TRACE_SWO
{35} MCU-PTA11 — 91 — PTA11/FTM1_CH5/FXIO_D1/CMP0_RRT
{35} MCU-PTA12 — 90 — PTA12/FTM1_CH6/CAN1_RX/FTM2_QD_PHB
{35} MCU-PTA13 — 89 — PTA13/FTM1_CH7/CAN1_TX/FTM2_QD_PHA
{35} MCU-PTA14 — 88 — PTA14/FTM0_FLT0/FTM3_FLT1/EWM_IN/FTM1_FLT0
{35} MCU-PTA15 — 83 — PTA15/FTM1_CH2/LPSPI0_PCS3/LPSPI2_PCS3/ADC1_SE12
{30} ADC1-SE13 — ADC1-SE13 — 82 — PTA16/FTM1_CH3/LPSPI1_PCS2/ADC1_SE13
62 — PTA17/FTM0_CH6/FTM3_FLT0/EWM_OUT_b

4 — PTD0/FTM0_CH2/LPSPI1_SCK/FTM2_CH0/FXIO_D0/TRGMUX_OUT1 — MCU-PTD0 {35}
3 — PTD1/FTM0_CH3/LPSPI1_SIN/FTM2_CH1/FXIO_D1/TRGMUX_OUT2 — MCU-PTD1 {35}
71 — R8221 33R — PTD2/FTM3_CH4/LPSPI1_SOUT/FXIO_D4/FXIO_D6/TRGMUX_IN5/ADC1_SE2
70 — R8222 0R — PTD3/FTM3_CH5/LPSPI1_PCS0/FXIO_D5/FXIO_D7/TRGMUX_IN4/NMI_b/ADC1_SE3 — LPSPI1-SOUT {34} / LPSPI1-PCS0 {34}
69 — PTD4/FTM0_FLT3/FTM3_FLT3/ADC1_SE6
33 — PTD5/FTM2_CH3/LPTMR0_ALT2/FTM2_FLT1/TRGMUX_IN7
32 — LPUART2-RX — PTD6/LPUART2_RX/FTM2_FLT2/CMP0_IN7 — LPUART2-RX {34}
31 — LPUART2-TX — PTD7/LPUART2_TX/FTM2_FLT3/CMP0_IN6 — LPUART2-TX {34}
42 — PTD8/FTM2_FLT2/FXIO_D1/FTM1_CH4 — MCU-PTD8 {35}
41 — PTD9/FXIO_D0/FTM2_FLT3/FTM1_CH5 — MCU-PTD9 {35}
36 — PTD10/FTM2_CH0/FTM2_QD_PHB — MCU-PTD10 {35}
35 — PTD11/FTM2_CH1/FTM2_QD_PHA/LPUART2_CTS — MCU-PTD11 {35}
34 — PTD12/FTM2_CH2/LPUART2_RTS — MCU-PTD12 {35}
25 — MCU-PTD13 — PTD13/FTM2_CH4/LPUART1_RX/RTC_CLKOUT — MCU-PTD13 {35}
24 — MCU-PTD14 — PTD14/FTM2_CH5/LPUART1_TX/CLKOUT — MCU-PTD14 {35}
22 — MCU-PTD15 — PTD15/FTM0_CH0/LPSPI0_SCK — MCU-PTD15 {35}
21 — MCU-PTD16 — PTD16/FTM0_CH1/LPSPI0_SIN/CMP0_RRT — MCU-PTD16 {35}
20 — SBC-LIN-RX — PTD17/FTM0_FLT2/LPUART2_RX — SBC-LIN-RX {34}

{36} ADC0-SE4 — ADC0-SE4 — 54 — PTB0/LPUART0_RX/LPSPI0_PCS0/LPTMR0_ALT3/CAN0_RX/ADC0_SE4/ADC1_SE14
{36} ADC0-SE5 — ADC0-SE5 — 53 — PTB1/LPUART0_TX/LPSPI0_SOUT/TCLK0/CAN0_TX/ADC0_SE5/ADC1_SE15
{36} ADC0-SE6 — ADC0-SE6 — 48 — PTB2/FTM1_CH0/LPSPI0_SCK/FTM1_QD_PHB/TRGMUX_IN3/ADC0_SE6
{36} ADC0-SE7 — ADC0-SE7 — 47 — PTB3/FTM1_CH1/LPSPI0_SIN/FTM1_QD_PHA/TRGMUX_IN2/ADC0_SE7
28 — PTB4/FTM0_CH4/LPSPI0_SOUT/TRGMUX_IN1
{34} SBC-LPSPI0_PCS0 — R8225 0R — 27 — PTB5/FTM0_CH5/LPSPI0_PCS1/LPSPI0_PCS0/CLKOUT/TRGMUX_IN0
XTAL-OUT — 16 — PTB6/LPI2C0_SDA/XTAL
XTAL-IN — 15 — PTB7/LPI2C0_SCL/EXTAL
77 — PTB8/FTM3_CH0
{35} MCU-PTB9 — MCU-PTB9 — 76 — PTB9/FTM3_CH1/LPI2C0_SCLS
{35} MCU-PTB10 — MCU-PTB10 — 75 — PTB10/FTM3_CH2/LPI2C0_SDAS
{35} MCU-PTB11 — 74 — PTB11/FTM3_CH3/LPI2C0_HREQ
{35} MCU-PTB12 — 68 — PTB12/FTM0_CH0/FTM3_FLT2/CAN2_RX/ADC1_SE7
{36} ADC0-SE8 — ADC0-SE8 — 67 — PTB13/FTM0_CH1/FTM3_FLT1/CAN2_TX/ADC1_SE8/ADC0_SE8
{34} LPSPI1-SCK — 33R R8219 — 66 — PTB14/FTM0_CH2/LPSPI1_SCK/ADC1_SE9/ADC0_SE9
{34} LPSPI1-SIN — 0R R8220 — 65 — PTB15/FTM0_CH3/LPSPI1_SIN/ADC1_SE14
64 — PTB16/FTM0_CH4/LPSPI1_SOUT/ADC1_SE15
63 — PTB17/FTM0_CH5/LPSPI1_PCS3

94 — R8218 33R — PTE0/LPSPI0_SCK/TCLK1/LPSPI1_SOUT/FTM1_FLT2 — SBC-LPSPI0-SCK {34}
93 — R8223 0R — PTE1/LPSPI0_SIN/LPI2C0_HREQ/LPSPI1_PCS0/FTM1_FLT1 — SBC-LPSPI0-SIN {34}
85 — R8224 33R — PTE2/LPSPI0_SOUT/LPTMR0_ALT3/FTM3_CH6/LPUART1_CTS/ADC1_SE10 — SBC-LPSPI0_SOUT {34}
18 — PTE3/FTM0_FLT0/LPUART2_RTS/FTM2_FLT0/TRGMUX_IN6/CMP0_OUT
9 — SBC-CAN0-RX — PTE4/FTM2_QD_PHB/FTM2_CH2/CAN0_RX/FXIO_D6/EWM_OUT_b — SBC-CAN0-RX {34}
8 — SBC-CAN0-TX — PTE5/TCLK2/FTM2_QD_PHA/FTM2_CH3/CAN0_TX/FXIO_D7/EWM_IN — SBC-CAN0-TX {34}
84 — PTE6/LPSPI0_PCS2/FTM3_CH7/LPUART1_RTS/ADC1_SE11 — MCU2PS-PWR-EN {37}
59 — PTE7/FTM0_CH7/FTM3_FLT0 — MCU-PTE7 {35}
26 — PTE8/FTM0_CH6/CMP0_IN3
23 — PTE9/FTM0_CH7/LPUART2_CTS
6 — MCU-PTE10 — PTE10/CLKOUT/LPSPI2_PCS1/FTM2_CH4/FXIO_D4/TRGMUX_OUT4 — MCU-PTE10 {35}
5 — MCU-LIN-SLP — PTE11/LPSPI2_PCS0/LPTMR0_ALT1/FTM2_CH5/FXIO_D5/TRGMUX_OUT5 — MCU-LIN-SLP {34}
19 — SBC-LIN-TX — PTE12/FTM0_FLT3/LPUART2_TX — SBC-LIN-TX {34}
7 — PTE13/LPSPI2_PCS2/FTM2_FLT0 — MCU-PTE13 {35}
17 — PTE14/FTM0_FLT1/FTM2_FLT1 — MCU-PTE14 {35}
2 — MCU-PTE15 — PTE15/LPUART1_CTS/LPSPI2_SCK/FTM2_CH6/FXIO_D2/TRGMUX_OUT6 — MCU-PTE15 {35}
1 — MCU-PTE16 — PTE16/LPUART1_RTS/LPSPI2_SIN/FTM2_CH7/FXIO_D3/TRGMUX_OUT7 — MCU-PTE16 {35}

{35} MCU-PTC0 — MCU-PTC0 — 40 — PTC0/FTM0_CH0/LPSPI2_SIN/FTM1_CH6/ADC0_SE8
{35} MCU-PTC1 — MCU-PTC1 — 39 — PTC1/FTM0_CH1/LPSPI2_SOUT/FTM1_CH7/ADC0_SE9
{35} MOTOR-SLEEP — 30 — PTC2/FTM0_CH2/CAN0_RX/LPUART0_RX/ADC0_SE10/CMP0_IN5
{35} MOTOR-ENABLE — 29 — PTC3/FTM0_CH3/CAN0_TX/LPUART0_TX/ADC0_SE11/CMP0_IN4
{36} SWD-CLK — SWD-CLK — 96 — PTC4/FTM1_CH0/RTC_CLKOUT/EWM_IN/FTM1_QD_PHB/JTAG_TCLK/SWD_CLK/CMP0_IN2
{35} MOTOR-RESET — 95 — PTC5/FTM2_CH0/RTC_CLKOUT/FTM2_QD_PHB/JTAG_TDI
{34} CAN1-RX — 81 — PTC6/LPUART1_RX/CAN1_RX/FTM3_CH2/FTM1_QD_PHB/ADC1_SE4
{34} CAN1-TX — 80 — PTC7/LPUART1_TX/CAN1_TX/FTM3_CH3/FTM1_QD_PHA/ADC1_SE5
{34} M-CAN1-STB — 56 — PTC8/LPUART1_RX/FTM1_FLT0/LPUART0_CTS
{35} MOTOR-DIR — 55 — PTC9/LPUART1_TX/FTM1_FLT1/LPUART0_RTS
{35} MOTOR_PWR_EN — 52 — PTC10/FTM3_CH4/TRGMUX_IN11
50 — PTC11/FTM3_CH5/TRGMUX_IN10
FTM2-CH7 — 49 — PTC12/FTM3_CH6/FTM2_CH6/LPUART2_CTS
{35} MOTOR-STEP — 46 — PTC13/FTM3_CH7/FTM2_CH7/LPUART2_RTS
{35} MCU-PTC14 — MCU-PTC14 — 45 — PTC14/FTM1_CH2/LPSPI2_PCS0/TRGMUX_IN9/ADC0_SE12
{35} MCU-PTC15 — MCU-PTC15 — 44 — PTC15/FTM1_CH3/LPSPI2_SCK/TRGMUX_IN8/ADC0_SE13
{35} MCU-PTC16 — 43 — PTC16/FTM1_FLT2/CAN2_RX/ADC0_SE14
{35} MCU-PTC17 — PTC17/FTM1_FLT3/CAN2_TX/ADC0_SE15

FS32K144UAT0VLLT

VDD5V-MCU
IDD < 100mA

10 — VDD_1
38 — VDD_2
61 — VDD_3
87 — VDD_4
C8176 10uF, C8177 1uF, C8178 100nF, C8179 100nF, C8180 100nF, C8181 100nF
11 — VDDA
12 — VREFH
13 — VREFL — R8285 0R
VDD5V-MCU
!ADC reference voltage
FB30 600R_100MHz 300mA
VREFH
C8182 10uF, C8183 100nF, C8184 1nF
14 — VSS_1
37 — VSS_2
60 — VSS_3
86 — VSS_4

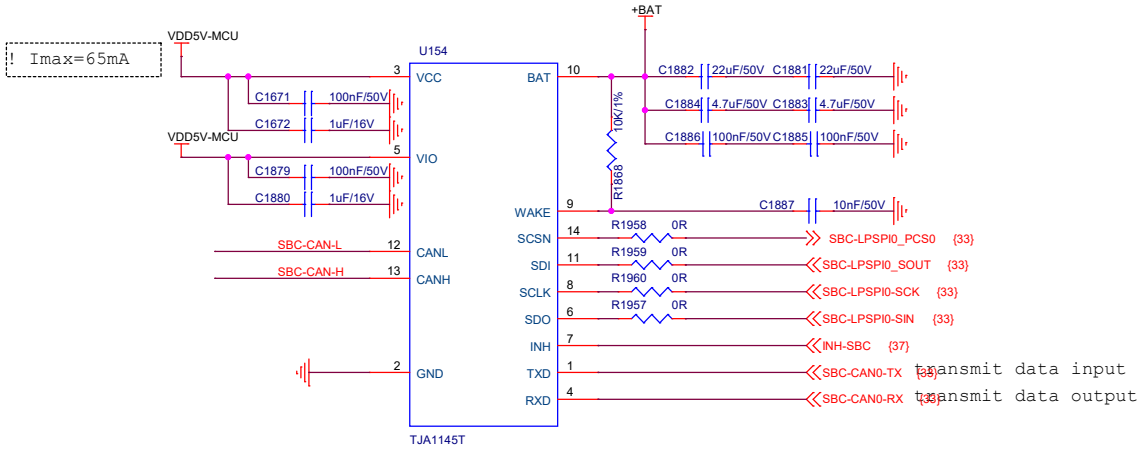
XTAL-IN — R1201 1M — XTAL-OUT
X7202 7U08000E12UCG 8MHz
GND1GND2
C1201 15pf
C1202 15pf
Fosc_hi =4~40MHz

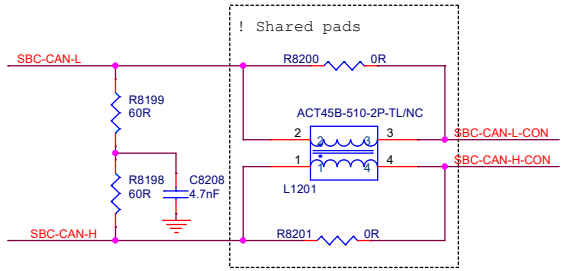
!MANUAL RESET FOR MCU
VDD5V-MCU
R1206 100K
MCU-RESET — R8217 100
TS-1088-AR02016
S6606
C1210 100nF
C8226 100NF
D8117 PESDMC2FD18VB

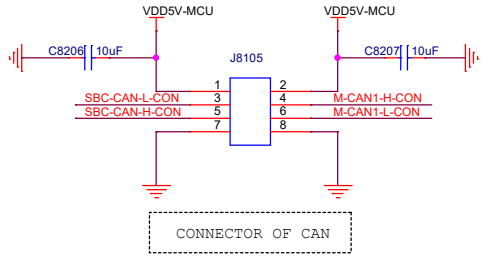
!NOTES
LSPI0 use for communication with SBC.
CAN0 use for communication with SBC.

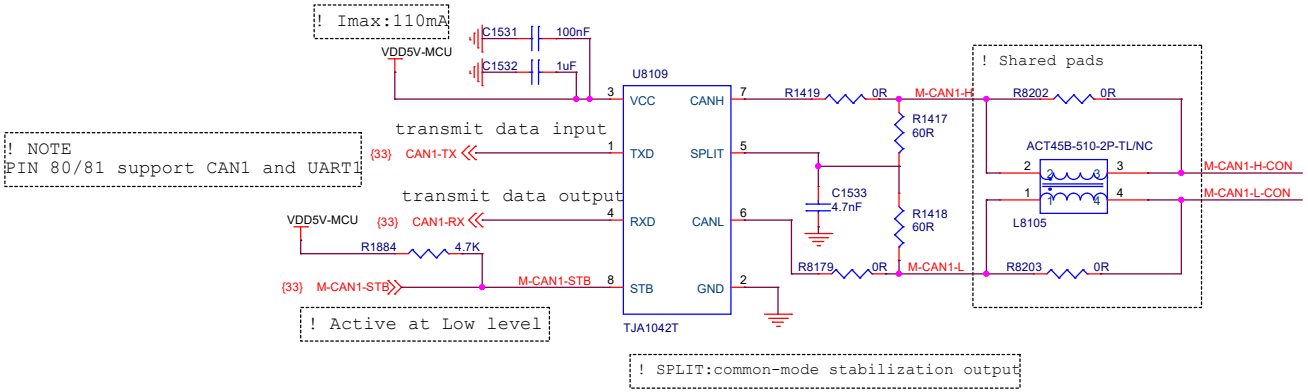
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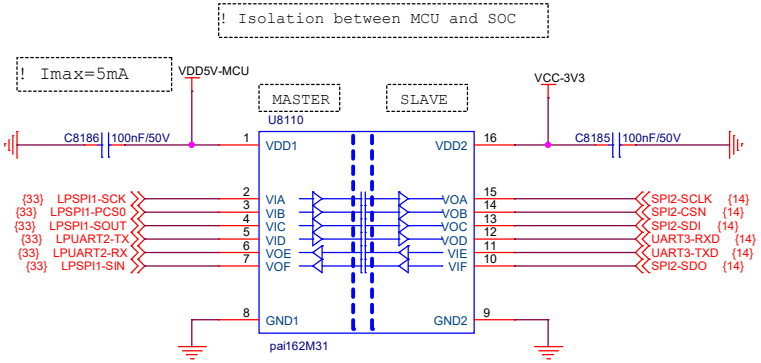
SBC&CAN

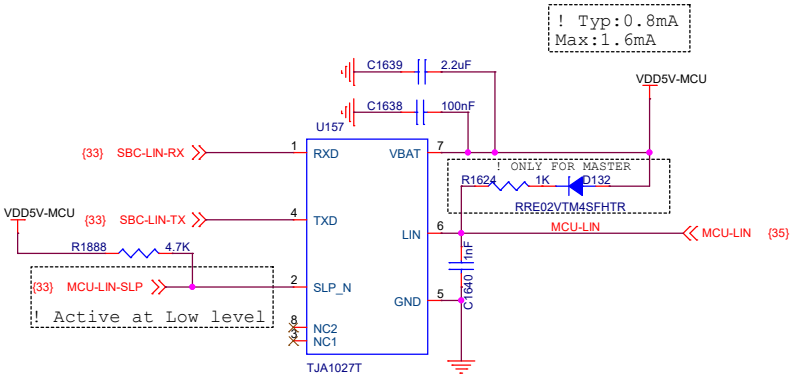






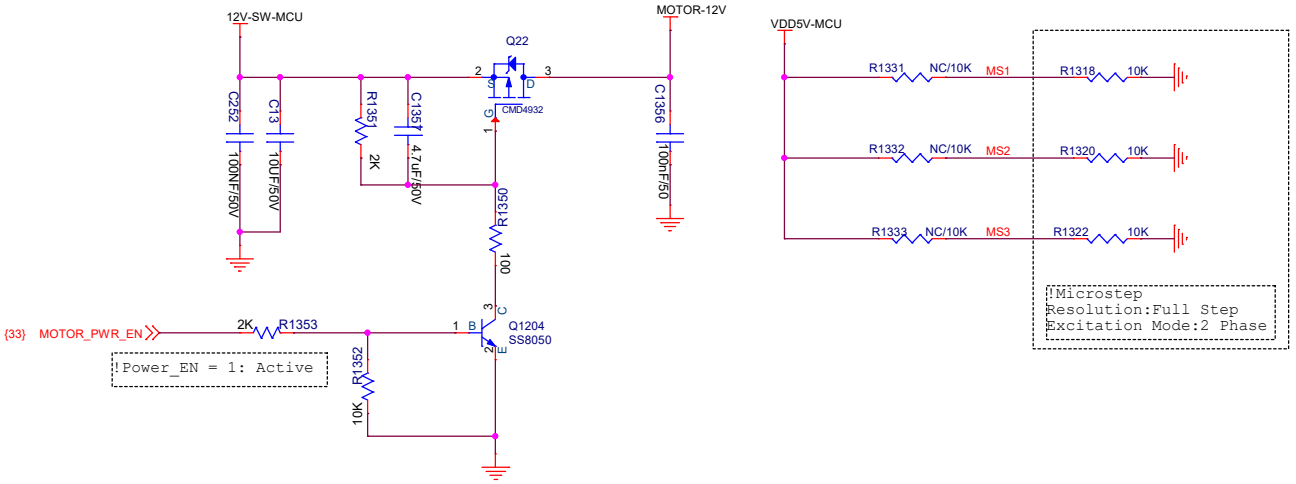




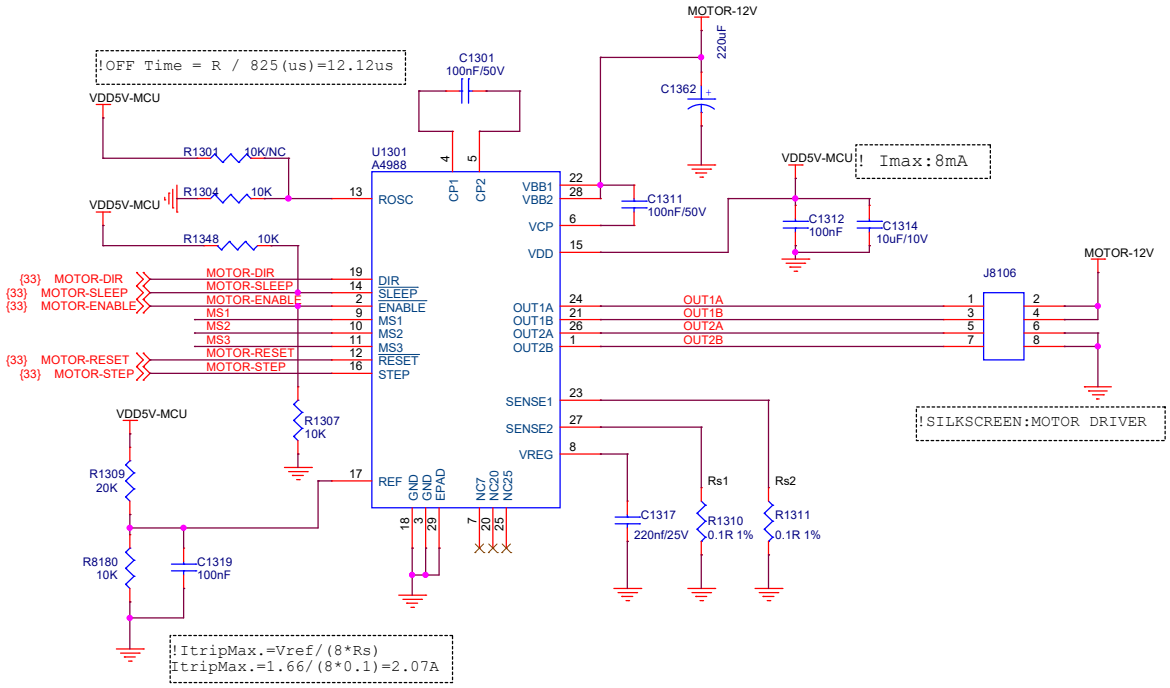


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Moter Driver

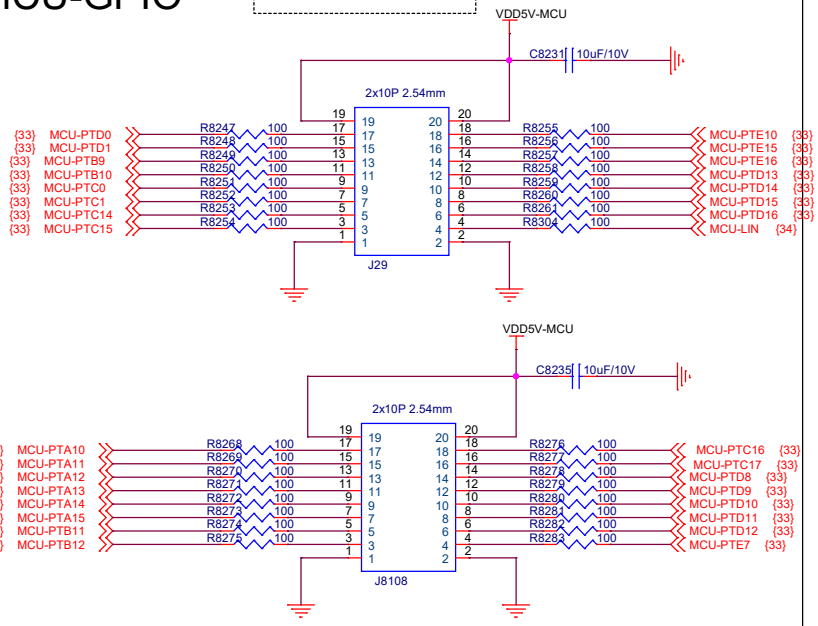


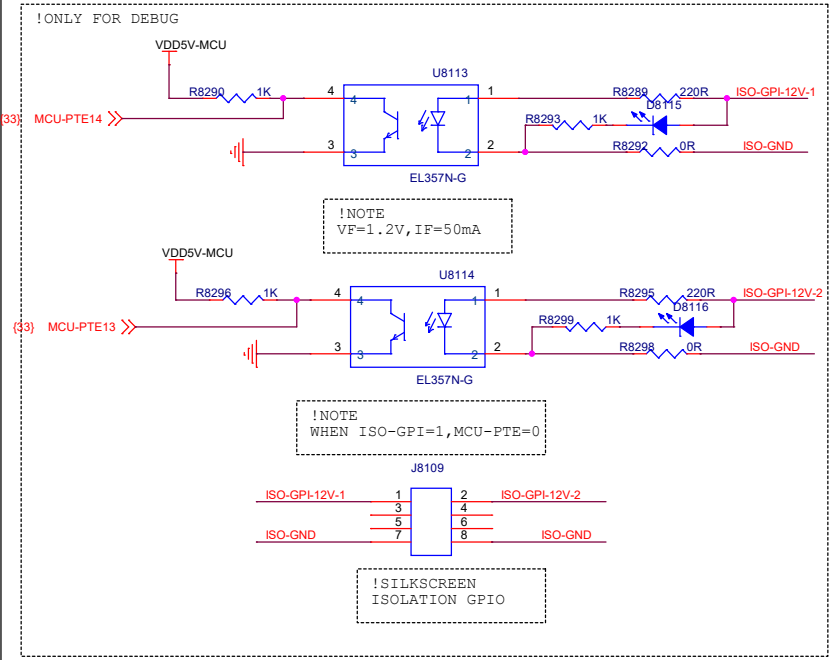
Maximum Rating 35V



MCU-GPIO

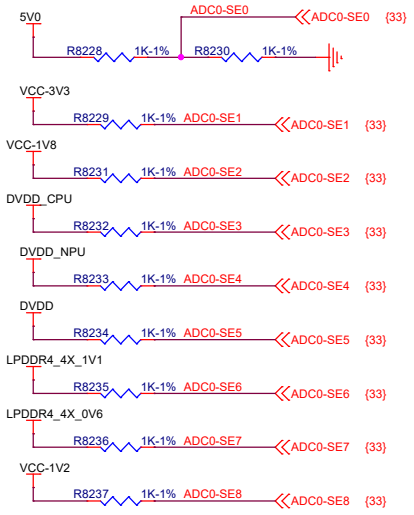
!NOTE
The IO voltage is 5V.





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!NOTES
SOC POWER DETECT



! (5V±10%)/2

! 3.135~3.465V

! 1.71~1.89V

! 0.65~1.05V

! 0.65~1.05V

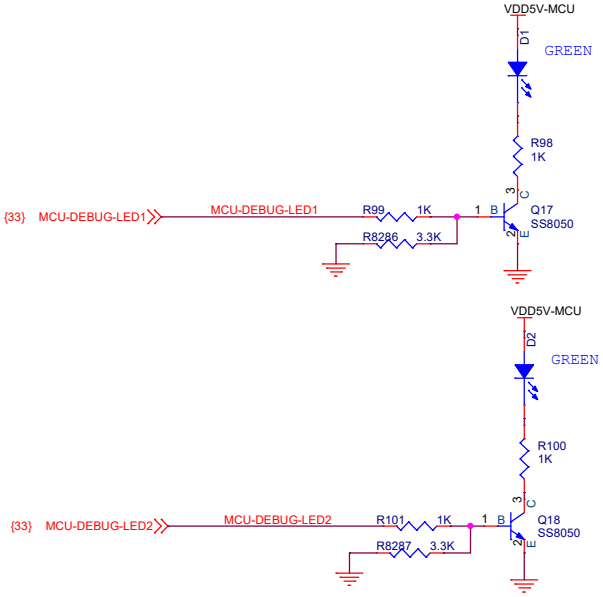
! 0.65~1.05V

! 1.045~1.155V

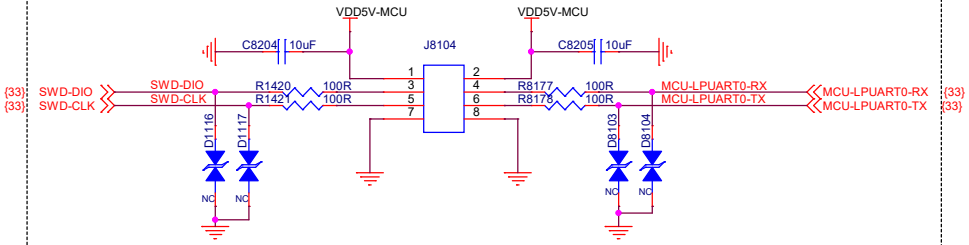
! 0.57~0.65V

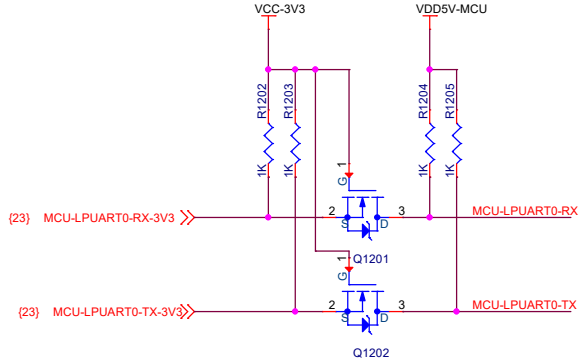
! 1.2V±5%

!JUST FOR DEBUG



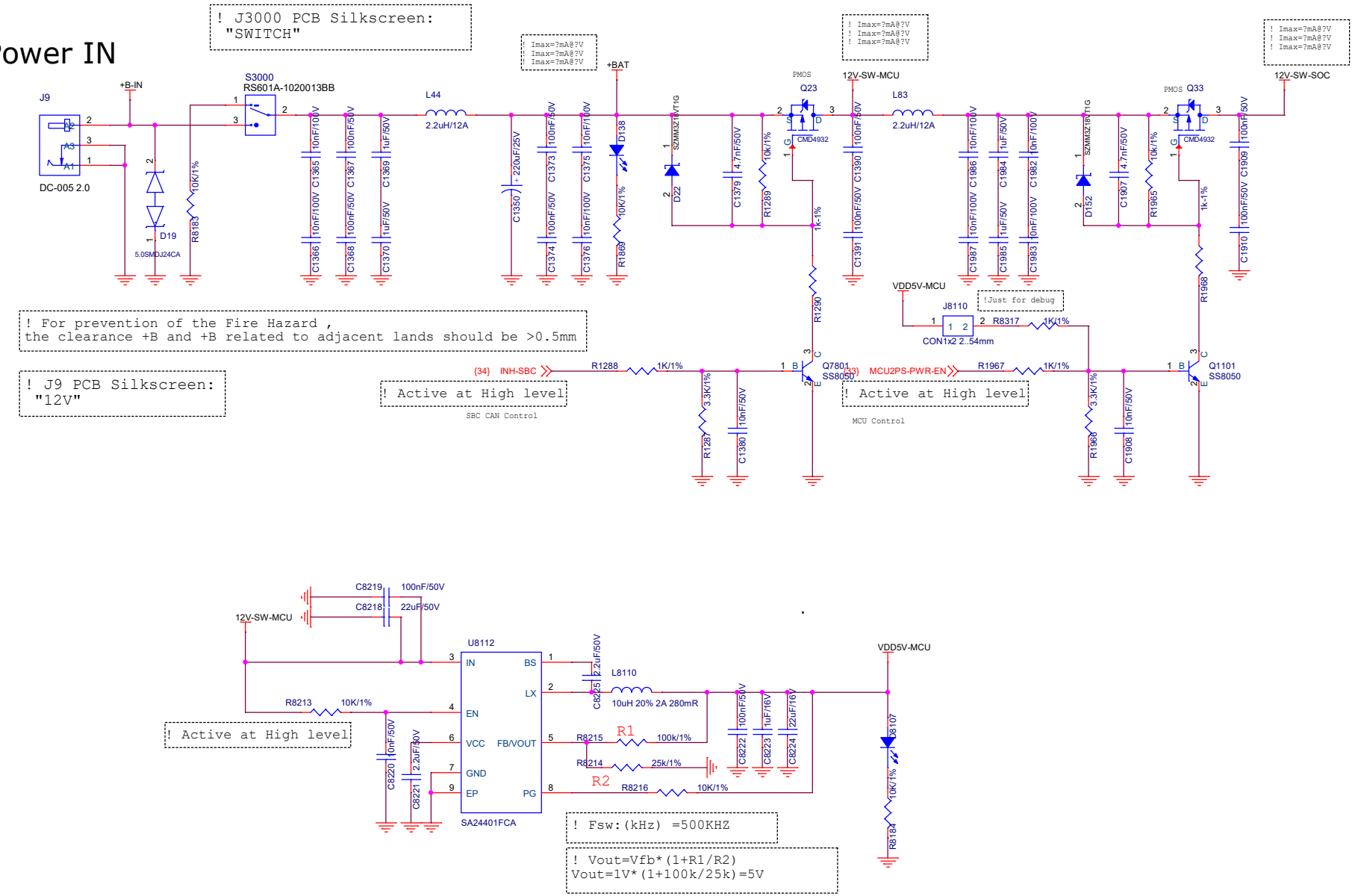
!JUST FOR DEBUG
SWD&UART





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Power IN



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